QUANTUM SECURITY INTELLIGENCE

Security & Post-Quantum Readiness Audit Report

jade @ a59d14c2b397 · Blockstream (Jade) — public repository (passive review)

Organization	QSI
Report date	2026-10-28 21:17 UTC
Report version	1
Audit run	633fed56-b826-4e1b-93cd-7d2ea40571e8 · Document audit · 2026-10-28 21:17 UTC
Analyzer engine	qsi-engine 1.2.0
Risk profile	Crypto exchange, custodian or wallet
Content SHA-256	`7942ddb55815a12070642368e6e3ea6fc9c02c06f4bad7fe1df427ff83cae3e3`
Signature key (Ed25519, SHA-256 fingerprint; the signature is the separate file report.sig.json)	`5b3cf9c9bd0fa54447e2b5dfa7a6bab569bd0c666fa46949599e5c348d17083e`

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This report presents automated and documented analysis results for the stated scope. It is not a legal opinion, a certification or an attestation of compliance with any standard.

Executive summary

8	0	0	7	0
Findings in total	Critical	High	Medium	Low

Classification	Count
Confirmed technical vulnerability	0
Known vulnerable component	0
Confirmed configuration error	0
Potential cryptographic risk	0
Long-term post-quantum risk	7
Improvement recommendation	1
Insufficient data	0

Top 5

ID	Severity	Detailed findings	Priority score
QSI-2026-06120	Medium	Quantum-vulnerable RSA key (4096-bit) in components/libwally-core/upstream/SECURITY.md	50.3
QSI-2026-06121	Medium	Quantum-vulnerable ECDSA: transaction and message signing (40 file(s)) preliminary	35.6
QSI-2026-06123	Medium	Quantum-vulnerable Schnorr (BIP-340): transaction and message signing (6 file(s)) preliminary	35.6
QSI-2026-06122	Medium	Quantum-vulnerable ECDSA: role not determined (2 file(s)) preliminary	34.2
QSI-2026-06124	Medium	Quantum-vulnerable ECDH: role not determined (1 file(s)) preliminary	34.2

Mosca's inequality (x + y > z): x = 10, y = 5, z = 8 → migration would finish too late under this scenario. z is a planning assumption chosen per risk profile, not a prediction. A cryptographically relevant quantum computer is not publicly known to exist today.

Post-quantum maturity: Level 0 of 4 — no post-quantum cryptography. Next step: Inventory where classical public-key cryptography protects long-lived data and keys, and choose the first migration: usually a hybrid TLS key exchange and wrapped keys.

Scope and boundaries

Mode: Document audit

Repository: https://github.com/Blockstream/Jade @ a59d14c2b39715060e6721a309a7d49bedd3ee5b — evidence links open the lines at this commit.

Supplied inputs

568 files, 10186070 bytes in total, 3 redaction(s); the per-file SHA-256 list is in appendix C.

Methodology

The audit was performed by the QSI engine 1.2.0. Each analyzer emits facts (rule, parameters, evidence); finding texts are generated from the rule catalog. Methodology sources are listed under "Sources and normative documents".

Prioritization uses an explainable weighted model (severity, exploitability, exposure, confidentiality/integrity/availability impact weighted by the "Crypto exchange, custodian or wallet" profile, asset criticality, urgency, ease of remediation) with a confidence multiplier. CVSS v3.1 is applied only to technical vulnerabilities with a defined vector. Long-term quantum risks use Mosca's inequality; their severity is capped at Medium and they are never presented as exploitable vulnerabilities.

Finding classes: confirmed technical vulnerability; known vulnerable component (a dependency version named in an advisory; whether the affected code is reachable is assessed separately, and a component confirmed by review becomes a vulnerability); confirmed configuration error; potential cryptographic risk; long-term post-quantum risk; improvement recommendation; insufficient data. Heuristic findings are marked preliminary; analyst decisions are shown next to the findings.

Coverage of checks

Area	Status	Count	Examples
source_code	checked	559	components/assets/assets_snapshot.c: 47 lines scanned with 16 rule patterns and 122 registry patterns; components/assets/assets_snapshot.h: 19 lines scanned with 16 rule patterns and 122 registry patterns; components/assets/gen_assets.py: 86 lines scanned with 16 rule patterns and 122 registry patterns
certificates_and_keys	checked	2	components/libwally-core/upstream/SECURITY.md: certificates/keys parsed; components/libwally-core/upstream/src/secp256k1/src/wycheproof/ecdsa_secp256k1_sha256_bitcoin_test.json: certificates/keys parsed
dependencies	checked	4	components/libwally-core/upstream/contrib/requirements.txt: 7 dependencies parsed (0 pinned); components/libwally-core/upstream/pyproject.toml: 0 dependencies parsed (0 pinned); components/libwally-core/upstream/src/wasm_package/package-lock.json: 156 dependencies parsed (156 pinned)
keys	error (not checked)	99	components/libwally-core/upstream/src/secp256k1/src/wycheproof/ecdsa_secp256k1_sha256_bitcoin_test.json line 134: public key could not be parsed (ValueError); components/libwally-core/upstream/src/secp256k1/src/wycheproof/ecdsa_secp256k1_sha256_bitcoin_test.json line 3600: public key could not be parsed (ValueError); components/libwally-core/upstream/src/secp256k1/src/wycheproof/ecdsa_secp256k1_sha256_bitcoin_test.json line 3636: public key could not be parsed (ValueError)
secrets	checked	3	components/libwally-core/upstream/src/swig_python/contrib/aes.py: value(s) at line(s) 12 (test_aes_cbc_ecdh), 15 (test_aes_cbc_ecdh) are declared inside a fixture and treated as test data, although the file itself is not test code; components/libwally-core/upstream/src/swig_python/contrib/aes.py: 2 redacted value(s) not reported (comments, test modules, fixture declarations, placeholders, values the source annotates as published, or hex values not bound to a secret name); libjade/selfcheck/descriptor.c: 1 redacted value(s) not reported (comments, test modules, fixture declarations, placeholders, values the source annotates as published, or hex values not bound to a secret name)

osv_advisories	checked	1	156 pinned packages queried at OSV.dev; 5 affected (5 of them only in development or build dependencies)

Long lists are grouped here; the full per-file list is in the JSON report, signed with the same key.

Assets examined and not examined

Supplied inputs: 566 (input files: C)

Not checked

- keys: 99 (examples: components/libwally-core/upstream/src/secp256k1/src/wycheproof/ecdsa_secp256k1_sha256_bitcoin_test.json line 134: public key could not be parsed (ValueError); components/libwally-core/upstream/src/secp256k1/src/wycheproof/ecdsa_secp256k1_sha256_bitcoin_test.json line 3600: public key could not be parsed (ValueError); components/libwally-core/upstream/src/secp256k1/src/wycheproof/ecdsa_secp256k1_sha256_bitcoin_test.json line 3636: public key could not be parsed (ValueError))

Architecture overview

Static review of the source tree jade at a59d14c2b39715060e6721a309a7d49bedd3ee5b. 568 files analysed; of 965 files, 159 are test, fuzzing, benchmark or example code and were left out, 238 are not read by any analyzer, 0 exceed the per-file limit and 0 did not fit the total limit. No requests were sent to the owner's systems.

Overall security posture

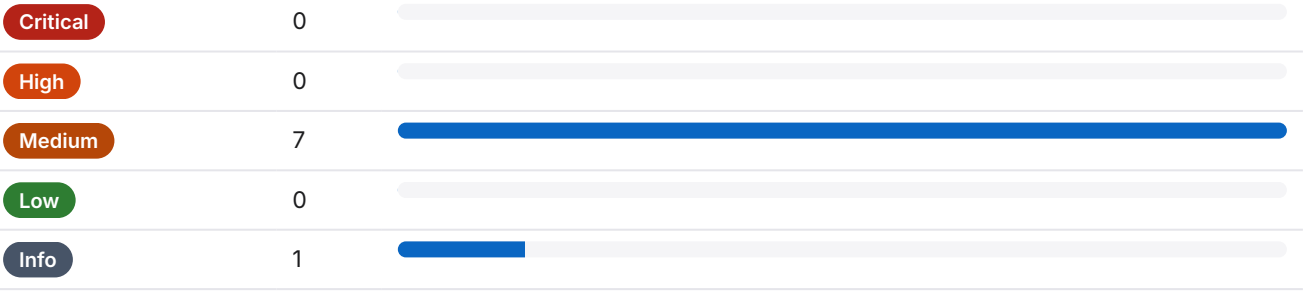
100
Indicative posture index (0–100)

71.0%
Quantum-vulnerable share of inventory

93
Cryptographic inventory

Derived from findings in scope (confirmed and potential issues weighted by severity). Indicative only; not comparable across differently scoped audits.

Findings by severity



ID	Detailed findings	Classification	Severity	Confidence
QSI-2026-06120	Quantum-vulnerable RSA key (4096-bit) in components/libwally-core/upstream/SECURITY.md	Long-term post-quantum risk	Medium	High
QSI-2026-06121	Quantum-vulnerable ECDSA: transaction and message signing (40 file(s))	Long-term post-quantum risk	Medium	Medium, preliminary
QSI-2026-06123	Quantum-vulnerable Schnorr (BIP-340): transaction and message signing (6 file(s))	Long-term post-quantum risk	Medium	Medium, preliminary
QSI-2026-06122	Quantum-vulnerable ECDSA: role not determined (2 file(s))	Long-term post-quantum risk	Medium	Medium, preliminary
QSI-2026-06124	Quantum-vulnerable ECDH: role not determined (1 file(s))	Long-term post-quantum risk	Medium	Medium, preliminary
QSI-2026-06125	Quantum-vulnerable Schnorr (BIP-340): role not determined (1 file(s))	Long-term post-quantum risk	Medium	Medium, preliminary
QSI-2026-06126	Quantum-vulnerable ECDSA: authentication and identity (certificates, tokens, device attestation) (1 file(s))	Long-term post-quantum risk	Medium	Medium, preliminary
QSI-2026-06127	15 advisories in 5 development or build-only npm package(s)	Improvement recommendation	Info	Medium, preliminary

Detailed findings

QSI-2026-06120 · KEY-QV-PUBLIC

Medium

Quantum-vulnerable RSA key (4096-bit) in components/libwally-core/upstream/SECURITY.md

Long-term post-quantum risk | Confidence: High

A RSA 4096-bit public key was found (line 52).

Affected asset	file:components/libwally-core/upstream/SECURITY.md#line 52
Data source	components/libwally-core/upstream/SECURITY.md
Detection method	Public key parsing.
Analyzer	certificates 1.2.0 · 2026-10-28 21:17 UTC
Priority score	50.3 / 100 (Crypto exchange, custodian or wallet)
CVSS v3.1	not applicable
Mosca's inequality (x + y > z)	x=10.0, y=5.0, z=8.0 → migration would finish too late under this scenario

POTENTIAL IMPACT

Long-term risk once a CRQC exists; for key exchange, recorded traffic could be decrypted retroactively.

REPRODUCTION STEPS

```
Open line 52 in the supplied file components/libwally-core/upstream/SECURITY.md and inspect the quoted lines (see evidence).
```

EVIDENCE

key · components/libwally-core/upstream/SECURITY.md · line 52 · SHA-256 18fb5f0b9bd19412… · File 3ab230590b464090…

```
OpenPGP primary key: algorithm ID 1 (RSA) 4096-bit
```

key · components/libwally-core/upstream/SECURITY.md · line 52 · SHA-256 484f2172c4dd53b2… · File 3ab230590b464090…

```
OpenPGP subkey: algorithm ID 1 (RSA) 4096-bit
```

LIMITATIONS OF THE CHECK

This is a forward-looking risk that depends on the arrival of a cryptographically relevant quantum computer (CRQC). No such computer is publicly known to exist today; the timeline is uncertain. It is not a present-day exploitable vulnerability.

RECOMMENDATION

Inventory and plan migration to ML-DSA (FIPS 204) or SLH-DSA (FIPS 205), in a hybrid/composite form where the ecosystem supports it..

RETEST PROCEDURE

Re-audit after migration.

P. W. Shor — Polynomial-Time Algorithms for Prime Factorization and Discrete Logarithms on a Quantum Computer (SIAM J. Comput., 1997); NIST IR 8547 — Transition to Post-Quantum Cryptography Standards

QSI-2026-06121 · CODE-QV-ASYMMETRIC

Medium

Quantum-vulnerable ECDSA: transaction and message signing (40 file(s))

Long-term post-quantum risk | Confidence: Medium | preliminary — requires further verification

146 reference(s) to ECDSA across 40 file(s), e.g. components/libwally-core/upstream/setup.py:58. Role, judged from paths, function names and nearby code: transaction and message signing. A CRQC derives the private key from a public key that is already known (on chain after the first spend, in reused addresses): funds and approvals controlled by these keys need a migration path.

Affected asset	source-code:ECDSA:transaction_signing
Data source	components/libwally-core/upstream/setup.py
Detection method	Regular-expression matching of crypto API patterns; the role comes from words in paths, enclosing function names and the lines around the matches.
Analyzer	source_code 1.2.0 · 2026-10-28 21:17 UTC
Priority score	35.6 / 100 (Crypto exchange, custodian or wallet)
CVSS v3.1	not applicable
Mosca's inequality (x + y > z)	x=10.0, y=5.0, z=8.0 → migration would finish too late under this scenario

POTENTIAL IMPACT

Long-term risk depends on what the algorithm protects: signatures (authenticity until migration) or key exchange/encryption (retroactive decryption of recorded data).

REPRODUCTION STEPS

```
Open components/libwally-core/upstream/setup.py:58 and inspect the quoted lines; the evidence lists
further files.
```

EVIDENCE

file_excerpt · components/libwally-core/upstream/setup.py · line 58 · SHA-256 faf6a0cba5f30b00… · File 016931c29406077b…

```
   57 |
   58 | if (USE_LIB == 'no' or not PKG_SECP256K1) and not os.path.exists('src/secp256k1/
Makefile.am'):
   59 |     # Sync libsecp256k1-zkp (only if needed)
```

file_excerpt · components/libwally-core/upstream/setup.py · line 75 · SHA-256 07224e4412156888… · File 016931c29406077b…

```
   74 | if PKG_SECP256K1:
   75 |     CONFIGURE_ARGS += ['--with-system-secp256k1=' + PKG_SECP256K1]
   76 |
```

file_excerpt · components/libwally-core/upstream/setup.py · line 132 · SHA-256 790764e66fa46e6f… · File 016931c29406077b…

```
  131 |         './src/ccan',
  132 |         './src/secp256k1/include',
  133 |     ]
```

file_excerpt · components/libwally-core/upstream/src/anti_exfil.c · line 72 · SHA-256 38a6257c59bd1576… · File 9a82b2e8a8fb25df…

```
   71 | {
   72 |     secp256k1_ecdsa_signature sig_secp;
   73 |     const secp256k1_context *ctx = secp_ctx();
```

file_excerpt · components/libwally-core/upstream/src/anti_exfil.c · line 87 · SHA-256 9213bc4125f159f4… · File 9a82b2e8a8fb25df…

```
   86 |     ok = secp256k1_anti_exfil_sign(ctx, &sig_secp, bytes, priv_key, entropy) &&
   87 |          secp256k1_ecdsa_signature_serialize_compact(ctx, bytes_out, &sig_secp);
   88 |
```

file_excerpt · components/libwally-core/upstream/src/anti_exfil.c · line 107 · SHA-256 ccc27036c049ec2f… · File 9a82b2e8a8fb25df…

```
  106 |     secp256k1_pubkey pub_secp;
  107 |     secp256k1_ecdsa_signature sig_secp;
  108 |     secp256k1_ecdsa_s2c_opening opening_secp;
```

file_excerpt · components/libwally-core/upstream/src/secp256k1/contrib/lax_der_parsing.c · line 11 · SHA-256 9197354c850cb53b… · File 4da2ecdc0a3b9dbb…

```
   10 |
   11 | int ecdsa_signature_parse_der_lax(const secp256k1_context* ctx, secp256k1_ecdsa_signature*
sig, const unsigned char *input, size_t inputlen) {
   12 |     size_t rpos, rlen, spos, slen;
```

file_excerpt · components/libwally-core/upstream/src/secp256k1/contrib/lax_der_parsing.c · line 19 · SHA-256 234aeaaf6b28e1d2… · File 4da2ecdc0a3b9dbb…

```
   18 |     /* Hack to initialize sig with a correctly-parsed but invalid signature. */
   19 |     secp256k1_ecdsa_signature_parse_compact(ctx, sig, tmpsig);
   20 |
```

file_excerpt · components/libwally-core/upstream/src/secp256k1/contrib/lax_der_parsing.c · line 140 · SHA-256 75ad5d0cba25ca64… · File 4da2ecdc0a3b9dbb…

```
  139 |     if (!overflow) {
  140 |         overflow = !secp256k1_ecdsa_signature_parse_compact(ctx, sig, tmpsig);
  141 |     }
```

file_excerpt · components/libwally-core/upstream/src/secp256k1/contrib/lax_der_parsing.h · line 57 · SHA-256 25af7e242d92af0d… · File 7f3e8ba009f19a5f…

```
   56 |    files (such as tests.c) without the need to set an explicit -I flag,
   57 |    which would be necessary to locate secp256k1.h. */
   58 | #ifndef SECP256K1_H
```

LIMITATIONS OF THE CHECK

This is a forward-looking risk that depends on the arrival of a cryptographically relevant quantum computer (CRQC). No such computer is publicly known to exist today; the timeline is uncertain. It is not a present-day exploitable vulnerability. Heuristic pattern match on source text without data-flow analysis. The usage context (security-relevant or not) must be confirmed manually; treat as preliminary.

RECOMMENDATION

Follow each chain's post-quantum roadmap, avoid public-key reuse, and keep signers ready to add post-quantum signature schemes as chains adopt them.

RETEST PROCEDURE

Re-run the inventory after refactoring.

P. W. Shor — Polynomial-Time Algorithms for Prime Factorization and Discrete Logarithms on a Quantum Computer (SIAM J. Comput., 1997); NIST IR 8547 — Transition to Post-Quantum Cryptography Standards; M. Mosca — Cybersecurity in an Era with Quantum Computers: Will We Be Ready? (IEEE Security & Privacy, 2018)

QSI-2026-06123 · CODE-QV-ASYMMETRIC

Medium

Quantum-vulnerable Schnorr (BIP-340): transaction and message signing (6 file(s))

Long-term post-quantum risk | Confidence: Medium | preliminary — requires further verification

37 reference(s) to Schnorr (BIP-340) across 6 file(s), e.g. components/libwally-core/upstream/src/secp256k1/include/secp256k1_schnorrsig.h:82. Role, judged from paths, function names and nearby code: transaction and message signing. A CRQC derives the private key from a public key that is already known (on chain after the first spend, in reused addresses): funds and approvals controlled by these keys need a migration path.

Affected asset	source-code:Schnorr (BIP-340):transaction_signing
Data source	components/libwally-core/upstream/src/secp256k1/include/secp256k1_schnorrsig.h
Detection method	Regular-expression matching of crypto API patterns; the role comes from words in paths, enclosing function names and the lines around the matches.
Analyzer	source_code 1.2.0 · 2026-10-28 21:17 UTC
Priority score	35.6 / 100 (Crypto exchange, custodian or wallet)
CVSS v3.1	not applicable
Mosca's inequality (x + y > z)	x=10.0, y=5.0, z=8.0 → migration would finish too late under this scenario

POTENTIAL IMPACT

Long-term risk depends on what the algorithm protects: signatures (authenticity until migration) or key exchange/encryption (retroactive decryption of recorded data).

REPRODUCTION STEPS

```
Open components/libwally-core/upstream/src/secp256k1/include/secp256k1_schnorrsig.h:82 and inspect the quoted lines; the evidence lists further files.
```

EVIDENCE

file_excerpt · components/libwally-core/upstream/src/secp256k1/include/secp256k1_schnorrsig.h · line 82 · SHA-256 6aae3b20f8f8b503… · File 8185d6c6ea3234ff…

```
 81 |  */
 82 | typedef struct secp256k1_schnorrsig_extraparams {
 83 |     unsigned char magic[4];
```

file_excerpt · components/libwally-core/upstream/src/secp256k1/include/secp256k1_schnorrsig.h · line 86 · SHA-256 43d5f6045ac4e6b2… · File 8185d6c6ea3234ff…

```
 85 |     void *ndata;
 86 | } secp256k1_schnorrsig_extraparams;
 87 |
```

file_excerpt · components/libwally-core/upstream/src/secp256k1/include/secp256k1_schnorrsig.h · line 119 · SHA-256 1e555d327421285c… · File 8185d6c6ea3234ff…

```
118 |  */
119 | SECP256K1_API int secp256k1_schnorrsig_sign32(
120 |     const secp256k1_context *ctx,
```

file_excerpt · components/libwally-core/upstream/src/secp256k1/include/secp256k1_schnorrsig_halfagg.h · line 43 · SHA-256 b3d54ae6f144846b… · File 89669a3ad2c90407…

```
   42 |  */
   43 | SECP256K1_API int secp256k1_schnorrsig_inc_aggregate(
   44 |     const secp256k1_context *ctx,
```

file_excerpt · components/libwally-core/upstream/src/secp256k1/include/secp256k1_schnorrsig_halfagg.h · line 70 · SHA-256 be378be4953e3b3d… · File 89669a3ad2c90407…

```
   69 |  */
   70 | SECP256K1_API int secp256k1_schnorrsig_aggregate(
   71 |     const secp256k1_context *ctx,
```

file_excerpt · components/libwally-core/upstream/src/secp256k1/include/secp256k1_schnorrsig_halfagg.h · line 94 · SHA-256 b10de912dd2b9f3e… · File 89669a3ad2c90407…

```
   93 |  */
   94 | SECP256K1_API SECP256K1_WARN_UNUSED_RESULT int secp256k1_schnorrsig_aggverify(
   95 |     const secp256k1_context *ctx,
```

file_excerpt · components/libwally-core/upstream/src/secp256k1/src/modules/musig/session_impl.h · line 621 · SHA-256 6254128e9a575e14… · File ddcb0d007ebaa3cc…

```
  620 |     secp256k1_musig_nonce_process_internal(ctx, &session_i.fin_nonce_parity, fin_nonce,
&session_i.noncecoef, aggnonce_pts, agg_pk32, msg32);
  621 |     secp256k1_schnorrsig_challenge(secp256k1_get_hash_context(ctx), &session_i.challenge,
fin_nonce, msg32, 32, agg_pk32);
  622 |
```

file_excerpt · components/libwally-core/upstream/src/secp256k1/src/modules/schnorrsig/bench_impl.h · line 33 · SHA-256 fc2c5559f32fc9bd… · File 9655992c7fd21423…

```
   32 |         msg[1] = i >> 8;
   33 |         CHECK(secp256k1_schnorrsig_sign_custom(data->ctx, sig, msg, MSGLEN, data-
>keypairs[i], NULL));
   34 |     }
```

file_excerpt · components/libwally-core/upstream/src/secp256k1/src/modules/schnorrsig/bench_impl.h · line 44 · SHA-256 4f2e9b7b2529431d… · File 9655992c7fd21423…

```
   43 |         CHECK(secp256k1_xonly_pubkey_parse(data->ctx, &pk, data->pk[i]) == 1);
   44 |         CHECK(secp256k1_schnorrsig_verify(data->ctx, data->sigs[i], data->msgs[i], MSGLEN,
&pk));
   45 |     }
```

file_excerpt · components/libwally-core/upstream/src/secp256k1/src/modules/schnorrsig/bench_impl.h · line 80 · SHA-256 0ca8c9b515200145… · File 9655992c7fd21423…

```
   79 |         CHECK(secp256k1_keypair_create(data.ctx, keypair, sk));
   80 |         CHECK(secp256k1_schnorrsig_sign_custom(data.ctx, sig, msg, MSGLEN, keypair, NULL));
   81 |         CHECK(secp256k1_keypair_xonly_pub(data.ctx, &pk, NULL, keypair));
```

LIMITATIONS OF THE CHECK

This is a forward-looking risk that depends on the arrival of a cryptographically relevant quantum computer (CRQC). No such computer is publicly known to exist today; the timeline is uncertain. It is not a present-day exploitable vulnerability. Heuristic pattern match on source text without data-flow analysis. The usage context (security-relevant or not) must be confirmed manually; treat as preliminary.

RECOMMENDATION

Follow each chain's post-quantum roadmap, avoid public-key reuse, and keep signers ready to add post-quantum signature schemes as chains adopt them.

RETEST PROCEDURE

Re-run the inventory after refactoring.

P. W. Shor — Polynomial-Time Algorithms for Prime Factorization and Discrete Logarithms on a Quantum Computer (SIAM J. Comput., 1997); NIST IR 8547 — Transition to Post-Quantum Cryptography Standards; M. Mosca — Cybersecurity in an Era with Quantum Computers: Will We Be Ready? (IEEE Security & Privacy, 2018)

QSI-2026-06122 · CODE-QV-ASYMMETRIC

Medium

Quantum-vulnerable ECDSA: role not determined (2 file(s))

Long-term post-quantum risk | Confidence: Medium | preliminary — requires further verification

4 reference(s) to ECDSA across 2 file(s), e.g. components/libwally-core/upstream/src/amalgamation/combined.c:34. Role, judged from paths, function names and nearby code: role not determined. The code does not show what the keys protect; review decides the priority.

Affected asset	source-code:ECDSA:general
Data source	components/libwally-core/upstream/src/amalgamation/combined.c
Detection method	Regular-expression matching of crypto API patterns; the role comes from words in paths, enclosing function names and the lines around the matches.
Analyzer	source_code 1.2.0 · 2026-10-28 21:17 UTC
Priority score	34.2 / 100 (Crypto exchange, custodian or wallet)
CVSS v3.1	not applicable
Mosca's inequality (x + y > z)	x=10.0, y=5.0, z=8.0 → migration would finish too late under this scenario

POTENTIAL IMPACT

Long-term risk depends on what the algorithm protects: signatures (authenticity until migration) or key exchange/encryption (retroactive decryption of recorded data).

REPRODUCTION STEPS

```
Open components/libwally-core/upstream/src/amalgamation/combined.c:34 and inspect the quoted lines;
the evidence lists further files.
```

EVIDENCE

file_excerpt · components/libwally-core/upstream/src/amalgamation/combined.c · line 34 · SHA-256 8cce795a23b61e66… · File ea2c2393b4f2f218…

```
33 | #endif
34 | #include "src/secp256k1/src/secp256k1.c"
35 | #include "src/secp256k1/src/precomputed_ecmult_gen.c"
```

file_excerpt · components/libwally-core/upstream/src/amalgamation/combined.c · line 35 · SHA-256 b0ba42431dd53977… · File ea2c2393b4f2f218…

```
34 | #include "src/secp256k1/src/secp256k1.c"
35 | #include "src/secp256k1/src/precomputed_ecmult_gen.c"
36 | #include "src/secp256k1/src/precomputed_ecmult.c"
```

file_excerpt · components/libwally-core/upstream/src/amalgamation/combined.c · line 36 · SHA-256 9f2414696e322db7… · File ea2c2393b4f2f218…

```
35 | #include "src/secp256k1/src/precomputed_ecmult_gen.c"
36 | #include "src/secp256k1/src/precomputed_ecmult.c"
37 |
```

file_excerpt · components/libwally-core/upstream/src/internal.h · line 33 · SHA-256 a3cf3e2c758ea293... · File f21377ed9eba49d0...

```
32 |
33 | #include <secp256k1.h>
34 | #include <secp256k1_recovery.h>
```

LIMITATIONS OF THE CHECK

This is a forward-looking risk that depends on the arrival of a cryptographically relevant quantum computer (CRQC). No such computer is publicly known to exist today; the timeline is uncertain. It is not a present-day exploitable vulnerability. Heuristic pattern match on source text without data-flow analysis. The usage context (security-relevant or not) must be confirmed manually; treat as preliminary.

RECOMMENDATION

Classify each usage, record data lifetimes, and plan migration to ML-DSA (FIPS 204) or SLH-DSA (FIPS 205), in a hybrid/composite form where the ecosystem supports it; introduce an abstraction layer for crypto-agility.

RETEST PROCEDURE

Re-run the inventory after refactoring.

P. W. Shor — Polynomial-Time Algorithms for Prime Factorization and Discrete Logarithms on a Quantum Computer (SIAM J. Comput., 1997); NIST IR 8547 — Transition to Post-Quantum Cryptography Standards; M. Mosca — Cybersecurity in an Era with Quantum Computers: Will We Be Ready? (IEEE Security & Privacy, 2018)

QSI-2026-06124 · CODE-QV-ASYMMETRIC

Medium

Quantum-vulnerable ECDH: role not determined (1 file(s))

Long-term post-quantum risk | Confidence: Medium | preliminary — requires further verification

4 reference(s) to ECDH across 1 file(s), e.g. components/libwally-core/upstream/src/secp256k1/src/bench.c:24. Role, judged from paths, function names and nearby code: role not determined. The code does not show what the keys protect; review decides the priority.

Affected asset	source-code:ECDH:general
Data source	components/libwally-core/upstream/src/secp256k1/src/bench.c
Detection method	Regular-expression matching of crypto API patterns; the role comes from words in paths, enclosing function names and the lines around the matches.
Analyzer	source_code 1.2.0 · 2026-10-28 21:17 UTC
Priority score	34.2 / 100 (Crypto exchange, custodian or wallet)
CVSS v3.1	not applicable
Mosca's inequality (x + y > z)	x=10.0, y=5.0, z=8.0 → migration would finish too late under this scenario

POTENTIAL IMPACT

Long-term risk depends on what the algorithm protects: signatures (authenticity until migration) or key exchange/encryption (retroactive decryption of recorded data).

REPRODUCTION STEPS

```
Open components/libwally-core/upstream/src/secp256k1/src/bench.c:24 and inspect the quoted lines;
the evidence lists further files.
```

EVIDENCE

file_excerpt · components/libwally-core/upstream/src/secp256k1/src/bench.c · line 24 · SHA-256 6ec91ca104f465f4… · File 832edacf541809e3…

```
23 | #ifdef ENABLE_MODULE_ECDH
24 |     printf("    - ECDH key exchange (optional module)\n");
25 | #endif
```

file_excerpt · components/libwally-core/upstream/src/secp256k1/src/bench.c · line 54 · SHA-256 b21ecd8d53157d4f… · File 832edacf541809e3…

```
53 | #ifdef ENABLE_MODULE_ECDH
54 |     printf("    ecdh              : ECDH key exchange algorithm\n");
55 | #endif
```

file_excerpt · components/libwally-core/upstream/src/secp256k1/src/bench.c · line 68 · SHA-256 ae7392892e7a0840… · File 832edacf541809e3…

```
67 |     printf("    ellswift_keygen   : ElligatorSwift key generation\n");
68 |     printf("    ellswift_ecdh     : ECDH on ElligatorSwift keys\n");
69 | #endif
```

LIMITATIONS OF THE CHECK

This is a forward-looking risk that depends on the arrival of a cryptographically relevant quantum computer (CRQC). No such computer is publicly known to exist today; the timeline is uncertain. It is not a present-day exploitable vulnerability. Heuristic pattern match on source text without data-flow analysis. The usage context (security-relevant or not) must be confirmed manually; treat as preliminary.

RECOMMENDATION

Classify each usage, record data lifetimes, and plan migration to Hybrid key establishment with ML-KEM (FIPS 203), e.g. X25519MLKEM768 in TLS 1.3 or mlkem768x25519-sha256 in SSH; introduce an abstraction layer for crypto-agility.

RETEST PROCEDURE

Re-run the inventory after refactoring.

P. W. Shor — Polynomial-Time Algorithms for Prime Factorization and Discrete Logarithms on a Quantum Computer (SIAM J. Comput., 1997); NIST IR 8547 — Transition to Post-Quantum Cryptography Standards; M. Mosca — Cybersecurity in an Era with Quantum Computers: Will We Be Ready? (IEEE Security & Privacy, 2018)

QSI-2026-06125 · CODE-QV-ASYMMETRIC

Medium

Quantum-vulnerable Schnorr (BIP-340): role not determined (1 file(s))

Long-term post-quantum risk | Confidence: Medium | preliminary — requires further verification

2 reference(s) to Schnorr (BIP-340) across 1 file(s), e.g. components/libwally-core/upstream/src/sign.c:387. Role, judged from paths, function names and nearby code: role not determined. The code does not show what the keys protect; review decides the priority.

Affected asset	source-code:Schnorr (BIP-340):general

Data source	components/libwally-core/upstream/src/sign.c
Detection method	Regular-expression matching of crypto API patterns; the role comes from words in paths, enclosing function names and the lines around the matches.
Analyzer	source_code 1.2.0 · 2026-10-28 21:17 UTC
Priority score	34.2 / 100 (Crypto exchange, custodian or wallet)
CVSS v3.1	not applicable
Mosca's inequality (x + y > z)	x=10.0, y=5.0, z=8.0 → migration would finish too late under this scenario

POTENTIAL IMPACT

Long-term risk depends on what the algorithm protects: signatures (authenticity until migration) or key exchange/encryption (retroactive decryption of recorded data).

REPRODUCTION STEPS

```
Open components/libwally-core/upstream/src/sign.c:387 and inspect the quoted lines; the evidence lists further files.
```

EVIDENCE

file_excerpt · components/libwally-core/upstream/src/sign.c · line 387 · SHA-256 b957a7949930facd... · File 9e52718e9015f768...

```
386 |             ret = WALLY_EINVAL;
387 |         else if (!secp256k1_schnorrsig_sign32(ctx, bytes_out, bytes, &keypair, aux_rand))
388 |             ret = WALLY_ERROR;
```

file_excerpt · components/libwally-core/upstream/src/sign.c · line 456 · SHA-256 b2cc452c48e864fe... · File 9e52718e9015f768...

```
455 |         ok = xpubkey_parse(&xonly_pub, pub_key, pub_key_len);
456 |         ok = ok && secp256k1_schnorrsig_verify(ctx, sig, bytes, bytes_len, &xonly_pub);
457 |         wally_clear(&xonly_pub, sizeof(xonly_pub));
```

LIMITATIONS OF THE CHECK

This is a forward-looking risk that depends on the arrival of a cryptographically relevant quantum computer (CRQC). No such computer is publicly known to exist today; the timeline is uncertain. It is not a present-day exploitable vulnerability. Heuristic pattern match on source text without data-flow analysis. The usage context (security-relevant or not) must be confirmed manually; treat as preliminary.

RECOMMENDATION

Classify each usage, record data lifetimes, and plan migration to a post-quantum algorithm; introduce an abstraction layer for crypto-agility.

RETEST PROCEDURE

Re-run the inventory after refactoring.

P. W. Shor — Polynomial-Time Algorithms for Prime Factorization and Discrete Logarithms on a Quantum Computer (SIAM J. Comput., 1997); NIST IR 8547 — Transition to Post-Quantum Cryptography Standards; M. Mosca — Cybersecurity in an Era with Quantum Computers: Will We Be Ready? (IEEE Security & Privacy, 2018)

QSI-2026-06126 · CODE-QV-ASYMMETRIC

Medium

Quantum-vulnerable ECDSA: authentication and identity (certificates, tokens, device attestation) (1 file(s))

Long-term post-quantum risk | Confidence: Medium | preliminary — requires further verification

3 reference(s) to ECDSA across 1 file(s), e.g. main/identity.c:251. Role, judged from paths, function names and nearby code: authentication and identity (certificates, tokens, device attestation). Short-lived authentication stays safe until a CRQC exists; long-lived identities (device attestation, CA and SSH keys) need a plan earlier.

Affected asset	source-code:ECDSA:authentication
Data source	main/identity.c
Detection method	Regular-expression matching of crypto API patterns; the role comes from words in paths, enclosing function names and the lines around the matches.
Analyzer	source_code 1.2.0 · 2026-10-28 21:17 UTC
Priority score	34.2 / 100 (Crypto exchange, custodian or wallet)
CVSS v3.1	not applicable
Mosca's inequality (x + y > z)	x=10.0, y=5.0, z=8.0 → migration would finish too late under this scenario

POTENTIAL IMPACT

Long-term risk depends on what the algorithm protects: signatures (authenticity until migration) or key exchange/encryption (retroactive decryption of recorded data).

REPRODUCTION STEPS

```
Open main/identity.c:251 and inspect the quoted lines; the evidence lists further files.
```

EVIDENCE

file_excerpt · main/identity.c · line 251 · SHA-256 e4f065671df536c8… · File 61059fdd5caf7e82…

```
  250 |     // Use RFC6979 deterministic signatures
  251 |     const int ret = mbedtls_ecdsa_sign_det_ext(&keypair->MBEDTLS_PRIVATE(grp), pr, ps,
&keypair->MBEDTLS_PRIVATE(d),
  252 |         challenge, challenge_len, MBEDTLS_MD_SHA256, random_mbedtls_cb, NULL);
```

file_excerpt · main/identity.c · line 254 · SHA-256 35d28b1ee44fa915… · File 61059fdd5caf7e82…

```
  253 |     if (ret) {
  254 |         JADE_LOGE("mbedtls_ecdsa_sign_det_ext() failed, returned %d", ret);
  255 |         return false;
```

file_excerpt · main/identity.c · line 275 · SHA-256 039845d8802aeee9… · File 61059fdd5caf7e82…

```
  274 |     // Sanity check
  275 |     JADE_ZERO_VERIFY(mbedtls_ecdsa_verify(
  276 |         &keypair->MBEDTLS_PRIVATE(grp), challenge, challenge_len, &keypair-
>MBEDTLS_PRIVATE(Q), pr, ps));
```

LIMITATIONS OF THE CHECK

This is a forward-looking risk that depends on the arrival of a cryptographically relevant quantum computer (CRQC). No such computer is publicly known to exist today; the timeline is uncertain. It is not a present-day exploitable vulnerability. Heuristic pattern match on source text without data-flow analysis. The usage context (security-relevant or not) must be confirmed manually; treat as preliminary.

RECOMMENDATION

Keep credentials short-lived and plan ML-DSA certificates, tokens and keys as the ecosystem supports them.

RETEST PROCEDURE

Re-run the inventory after refactoring.

P. W. Shor — Polynomial-Time Algorithms for Prime Factorization and Discrete Logarithms on a Quantum Computer (SIAM J. Comput., 1997); NIST IR 8547 — Transition to Post-Quantum Cryptography Standards; M. Mosca — Cybersecurity in an Era with Quantum Computers: Will We Be Ready? (IEEE Security & Privacy, 2018)

QSI-2026-06127 · DEP-ADVISORY-DEV

Info

15 advisories in 5 development or build-only npm package(s)

Improvement recommendation | Confidence: Medium | preliminary — requires further verification

OSV.dev lists 15 advisories (8 rated high or critical) for 5 npm package(s) that only development or build tooling pulls in (components/libwally-core/upstream/src/wasm_package/package-lock.json): e.g. ajv@8.17.1 [dev]: GHSA-2g4f-4pwh-qvx6; baseline-browser-mapping@2.9.19 [dev]: GHSA-w5vr-8v7q-w6rv; browserslist@4.28.1 [dev]: GHSA-73wf-gq98-2v4g, GHSA-c83g-rgw3-j3cx; cross-spawn@7.0.3 [dev]: GHSA-3xgq-45jj-v275; fast-uri@3.1.5 [dev]: GHSA-333x-w4r6-6crp, GHSA-5jgf-p345-68v8, GHSA-f65p-4m7j-42xc, GHSA-fph4-wmhf-6fwf, GHSA-hrr3-gc8f-f4qj, GHSA-jqff-g426-hqxp, GHSA-qw65-cvwx-89v3, GHSA-w74x-v6f9-j3xg, GHSA-wjr9-j795-jxcr, GHSA-xhq7-xmqj-769h. The manifests mark them as such (devDependencies, "dev": true in package-lock.json, pnpm importers sections, [dev-dependencies]/[build-dependencies] reached through Cargo.lock, or — for yarn.lock, which records no scope at all — package.json reached through the lock graph), so they are not part of what ships.

Affected asset	dependencies:npm:non-runtime
Data source	OSV.dev
Detection method	OSV.dev querybatch lookup of exact versions; scope from devDependencies, package-lock.json dev flags, pnpm importers sections, the Cargo.lock graph rooted at the Cargo.toml tables, and — for yarn.lock, which records no scope — the lock graph walked from package.json. Where a runtime requirement cannot be resolved in a lock, that lock is left unclassified rather than risking a package being set aside wrongly.
Analyzer	dependencies 1.2.0 · 2026-10-28 21:17 UTC
Priority score	15.5 / 100 (Crypto exchange, custodian or wallet)
CVSS v3.1	not applicable

POTENTIAL IMPACT

They do not reach users of the product. They can affect developer machines and CI (supply chain): a compromised or vulnerable tool can tamper with builds.

REPRODUCTION STEPS

```
Look the packages listed in the evidence up at https://osv.dev and check their scope in components/
libwally-core/upstream/src/wasm_package/package-lock.json.
```

EVIDENCE

tool_output · OSV.dev (npm) · components/libwally-core/upstream/src/wasm_package/package-lock.json · SHA-256 33de8bddd802999e…

```
ajv@8.17.1 [dev]: GHSA-2g4f-4pwh-qvx6
baseline-browser-mapping@2.9.19 [dev]: GHSA-w5vr-8v7q-w6rv
browserslist@4.28.1 [dev]: GHSA-73wf-gq98-2v4g, GHSA-c83g-rgw3-j3cx
cross-spawn@7.0.3 [dev]: GHSA-3xgq-45jj-v275
fast-uri@3.1.5 [dev]: GHSA-333x-w4r6-6crp, GHSA-5jgf-p345-68v8, GHSA-f65p-4m7j-42xc, GHSA-fph4-
```

```
wmhf-6fwf, GHSA-hrr3-gc8f-f4qj, GHSA-jqff-g426-hqxp, GHSA-qw65-cvwx-89v3, GHSA-w74x-v6f9-j3xg, GHSA-
wjr9-j795-jxcr, GHSA-xhq7-xmqj-769h
```

LIMITATIONS OF THE CHECK

Advisory data comes from OSV.dev at the recorded time. Scopes come from the manifests; a registry crate's own build dependencies cannot be told apart in Cargo.lock, so the runtime set may be larger than reality, never smaller.

RECOMMENDATION

Update the tooling during routine maintenance; prioritise packages that run in CI with secrets or that produce release artefacts.

RETEST PROCEDURE

Re-run the dependency scan after updating.

OSV.dev — open-source vulnerability database and API

Evidence

ID	Type	Data source	SHA-256
QSI-2026-06120	key	components/libwally-core/upstream/SECURITY.md (line 52)	18fb5f0b9bd19412cb5c0404d29bd26dc62ba6e0ff99543463ebc
QSI-2026-06120	key	components/libwally-core/upstream/SECURITY.md (line 52)	484f2172c4dd53b29fd23e321304b308266950550fb37c8343147
QSI-2026-06121	file_excerpt	components/libwally-core/upstream/setup.py (line 58)	faf6a0cba5f30b003359e096b202e6312b83d259b06c8c18493fc
QSI-2026-06121	file_excerpt	components/libwally-core/upstream/setup.py (line 75)	07224e44121568888046b6db0a64d8c77d33d1715e2c4c0e03b44
QSI-2026-06121	file_excerpt	components/libwally-core/upstream/setup.py (line 132)	790764e66fa46e6f76061b300ae744943186e2a604de2f62962c3
QSI-2026-06121	file_excerpt	components/libwally-core/upstream/src/anti_exfil.c (line 72)	38a6257c59bd1576fcad77de26a0b8b07e8d637f0457829ebd386
QSI-2026-06121	file_excerpt	components/libwally-core/upstream/src/anti_exfil.c (line 87)	9213bc4125f159f48bc6b0e8d7906dc826e6431c9b90cc8d1e9b9
QSI-2026-06121	file_excerpt	components/libwally-core/upstream/src/anti_exfil.c (line 107)	ccc27036c049ec2fe55b8cab5992c66b4834971552c32740b52ae
QSI-2026-06121	file_excerpt	components/libwally-core/upstream/src/secp256k1/contrib/lax_der_parsing.c (line 11)	9197354c850cb53bbc475b348c0f97bb3f550706fcced8bf48333
QSI-2026-06121	file_excerpt	components/libwally-core/upstream/src/secp256k1/contrib/lax_der_parsing.c (line 19)	234aeaaf6b28e1d2ff43854e60a3119e3c3f55229e18841e2dc76
QSI-2026-06121	file_excerpt	components/libwally-core/upstream/src/secp256k1/contrib/lax_der_parsing.c (line 140)	75ad5d0cba25ca64adb6d971da937879704b4abc8845c17dc1da9
QSI-2026-06121	file_excerpt	components/libwally-core/upstream/src/secp256k1/contrib/lax_der_parsing.h (line 57)	25af7e242d92af0d3c46ae8d8a48030e247c4bfdeba65068030e6
QSI-2026-06123	file_excerpt	components/libwally-core/upstream/src/secp256k1/include/secp256k1_schnorrsig.h (line 82)	6aae3b20f8f8b503597acb97fe491246b6777b5a9660b8a415426

QSI-2026-06123	file_excerpt	components/libwally-core/upstream/src/secp256k1/include/secp256k1_schnorrsig.h (line 86)	43d5f6045ac4e6b23daf11fd6c10022dca16ca76afd775c3ee898
QSI-2026-06123	file_excerpt	components/libwally-core/upstream/src/secp256k1/include/secp256k1_schnorrsig.h (line 119)	1e555d327421285c8a81345e58876f5f2f017571adb2f2168aafc
QSI-2026-06123	file_excerpt	components/libwally-core/upstream/src/secp256k1/include/secp256k1_schnorrsig_halfagg.h (line 43)	b3d54ae6f144846b9769cfe141d006f33bf49fa9572bafb7307c2
QSI-2026-06123	file_excerpt	components/libwally-core/upstream/src/secp256k1/include/secp256k1_schnorrsig_halfagg.h (line 70)	be378be4953e3b3d2e2b5a4ddab26874bc43e6f72392495b730ca
QSI-2026-06123	file_excerpt	components/libwally-core/upstream/src/secp256k1/include/secp256k1_schnorrsig_halfagg.h (line 94)	b10de912dd2b9f3e510c8fe909f0b6db3a25cde95edbbe0af55b1
QSI-2026-06123	file_excerpt	components/libwally-core/upstream/src/secp256k1/src/modules/musig/session_impl.h (line 621)	6254128e9a575e14bb87c68d5f2f12d65a6084dd7de58385e4a27
QSI-2026-06123	file_excerpt	components/libwally-core/upstream/src/secp256k1/src/modules/schnorrsig/bench_impl.h (line 33)	fc2c5559f32fc9bd9a85efef59a46eaf722b1b8e2745b837696cd
QSI-2026-06123	file_excerpt	components/libwally-core/upstream/src/secp256k1/src/modules/schnorrsig/bench_impl.h (line 44)	4f2e9b7b2529431d8190d606bbe0ce49659d6e5a1e0410625d931
QSI-2026-06123	file_excerpt	components/libwally-core/upstream/src/secp256k1/src/modules/schnorrsig/bench_impl.h (line 80)	0ca8c9b51520014576b4c622e33014e8b93cc7c4bd0bc826e9cd4
QSI-2026-06122	file_excerpt	components/libwally-core/upstream/src/amalgamation/combined.c (line 34)	8cce795a23b61e665185923e4058d56ed7e106c55784ee042d2a2
QSI-2026-06122	file_excerpt	components/libwally-core/upstream/src/amalgamation/combined.c (line 35)	b0ba42431dd53977e931a49f059369c0a07f7bb7b7924b65b78a0

QSI-2026-06122	file_excerpt	components/libwally-core/upstream/src/amalgamation/combined.c (line 36)	9f2414696e322db70b097fe19907ea2f68a74d7db34037aad03b8
QSI-2026-06122	file_excerpt	components/libwally-core/upstream/src/internal.h (line 33)	a3cf3e2c758ea293394dc165dc129bd87d56cac6ce0aa935b84a4
QSI-2026-06124	file_excerpt	components/libwally-core/upstream/src/secp256k1/src/bench.c (line 24)	6ec91ca104f465f4b861193bee0ed1f6d91c4074fb5085511ca06
QSI-2026-06124	file_excerpt	components/libwally-core/upstream/src/secp256k1/src/bench.c (line 54)	b21ecd8d53157d4f920477f57713fa34b7fe6379df61daa64f27e
QSI-2026-06124	file_excerpt	components/libwally-core/upstream/src/secp256k1/src/bench.c (line 68)	ae7392892e7a0840c3596f3442ca87927d89cc694ec91267d2a73
QSI-2026-06125	file_excerpt	components/libwally-core/upstream/src/sign.c (line 387)	b957a7949930facdb75237e2e6c12333acd87496052f7b47d028b
QSI-2026-06125	file_excerpt	components/libwally-core/upstream/src/sign.c (line 456)	b2cc452c48e864fed16b916f1e68559e9d63deec6656b7f76f359
QSI-2026-06126	file_excerpt	main/identity.c (line 251)	e4f065671df536c8ac1aa4a99bc178b3d17ff42fa11fb09a36df3
QSI-2026-06126	file_excerpt	main/identity.c (line 254)	35d28b1ee44fa915a5b1e3e193e36c19d55df56f5841f6b478e1c
QSI-2026-06126	file_excerpt	main/identity.c (line 275)	039845d8802aeee967d9d03901cae13aa3efd9a0f0f29b8a4fa00
QSI-2026-06127	tool_output	OSV.dev (npm) (components/libwally-core/upstream/src/wasm_package/package-lock.json)	33de8bddd802999eabbfb4e4b5e3ecf4a6765d989bd5c6b7e1fbc

Post-quantum assessment

	Key establishment (x = how long data must stay confidential; harvest now, decrypt later)	Signatures (x = how long signatures/ keys must stay trusted)
Data/key lifetime x	10	10
Migration time y	5	5
Assumed years to CRQC z	8 (early)	8
x + y − z	7 → migration would finish too late under this scenario	7 → migration would finish too late under this scenario

Planning assumptions: x for key establishment, x for signatures, y are the defaults of the risk profile “Crypto exchange, custodian or wallet”, not the client's figures; with the client's actual lifetimes the verdict can change.

Post-quantum maturity: Level 0 of 4 — no post-quantum cryptography

Next step: Inventory where classical public-key cryptography protects long-lived data and keys, and choose the first migration: usually a hybrid TLS key exchange and wrapped keys.

Levels: 0 none; 1 present, not wired; 2 in use outside the critical paths; 3 hybrid in a critical path; 4 post-quantum by default in the critical paths (root of trust, key wrapping, channel key exchange). Judged from code and manifests.

What the quantum-vulnerable algorithms do

Role	Algorithms (files)	ID	First step
transaction and message signing	ECDSA (40), Schnorr (BIP-340) (6)	QSI-2026-06121, QSI-2026-06123	Follow each chain's post-quantum roadmap, avoid public-key reuse, and keep signers ready to add post-quantum signature schemes as chains adopt them
authentication and identity (certificates, tokens, device attestation)	ECDSA (1)	QSI-2026-06126	Keep credentials short-lived and plan ML-DSA certificates, tokens and keys as the ecosystem supports them
role not determined	ECDSA (2), ECDH (1), Schnorr (BIP-340) (1)	QSI-2026-06122, QSI-2026-06124, QSI-2026-06125	Classify each usage, record data lifetimes, and plan migration to ML-DSA (FIPS 204) or SLH-DSA (FIPS 205), in a hybrid/composite form where the ecosystem supports it; introduce an abstraction layer for crypto-agility

Roles are judged from file paths, function names and the code around each match; review confirms them. Root-of-trust keys and wrapped keys come first: they are hard to replace or can be recorded today.

z is a planning assumption chosen per risk profile, not a prediction. A cryptographically relevant quantum computer is not publicly known to exist today.

Quantum status	Count
weak classically	2
reduced margin (Grover), approved	19
Grover: acceptable to NIST, below ECCG/ANSSI guidance	6
vulnerable (Shor)	66

Detailed findings: QSI-2026-06120, QSI-2026-06121, QSI-2026-06123, QSI-2026-06122, QSI-2026-06124, QSI-2026-06125, QSI-2026-06126

PQC readiness questionnaire

No readiness questionnaire was completed for this project.

Quantum algorithms relevant to this inventory

Quantum algorithm	Effect	Inventoried algorithms (rows)	What it means for this audit
Shor's algorithm — factoring Q1: quantum computation offline, classical access to the victim	breaks	RSA (4)	Breaks RSA at every key length: a longer modulus only raises the cost polynomially. Plan migration to ML-KEM (or a hybrid) for key establishment and ML-DSA / SLH-DSA for signatures; start with keys protecting data that must stay confidential for years (harvest now, decrypt later) and with long-lived signatures.
Shor's algorithm — elliptic-curve discrete logarithms Q1: quantum computation offline, classical access to the victim	breaks	ECDH (3), ECDSA (52), Schnorr (BIP-340) (7)	Elliptic-curve keys are expected to fall before RSA keys of comparable classical strength. On blockchains, coins whose public key is already on chain (reused addresses, pay-to-public-key and Taproot outputs) are exposed first; plan the move to post-quantum signatures and avoid key reuse in the meantime.
Grover's algorithm — key search and preimages Q1: quantum computation offline, classical access to the victim	weakens (larger keys restore the margin)	HMAC (9), PBKDF2 (1), RIPEMD-160 (2), SHA-2 (8), SHAKE (2), scrypt (5)	Not a break. 256-bit keys (AES-256, ChaCha20) keep a comfortable margin; AES-128 meets NIST category 1, while some authorities ask for 256-bit keys in quantum-resistant contexts (see the AES-128 entry). Prefer 256-bit keys for long-lived data; 256-bit hashes need no change for preimage resistance.
Brassard–Høyer–Tapp — collision search Q1: quantum computation offline, classical access to the victim	weakens (larger keys restore the margin)	RIPEMD-160 (2), SHA-2 (8), SHAKE (2)	256-bit hashes (SHA-256, SHA3-256) remain adequate for collision resistance; 160-bit hashes are already weak classically (SHA-1 collisions are practical). Use 384/512-bit hashes where NIST categories 4–5 are required.

Effects: breaks — replace the algorithm; weakens — larger keys or outputs restore the margin; included in standard parameters — no action at standard parameter sets; only with superposition queries — not applicable to ordinary deployments.

Limitations and assumptions

- This report presents automated and documented analysis results for the stated scope. It is not a legal opinion, a certification or an attestation of compliance with any standard.
- z is a planning assumption chosen per risk profile, not a prediction. A cryptographically relevant quantum computer is not publicly known to exist today.
- keys — error (not checked): 99 (examples: components/libwally-core/upstream/src/secp256k1/src/wycheproof/ecdsa_secp256k1_sha256_bitcoin_test.json line 134: public key could not be parsed (ValueError); components/libwally-core/upstream/src/secp256k1/src/wycheproof/ecdsa_secp256k1_sha256_bitcoin_test.json line 3600: public key could not be parsed (ValueError); components/libwally-core/upstream/src/secp256k1/src/wycheproof/ecdsa_secp256k1_sha256_bitcoin_test.json line 3636: public key could not be parsed (ValueError))

Remediation plan

QSI-2026-06120 · P3 · 90 days short-term risk reduction

Add RSA usages at file:components/libwally-core/upstream/SECURITY.md to the cryptographic inventory with an owner and a data-lifetime estimate; design for crypto-agility (algorithm identifiers, configurable providers).

Rationale	Migration cannot be planned without knowing where and why the algorithm is used.
Expected outcome	Every usage has an owner, a lifetime and a migration path.
Complexity	medium
Dependencies	Access to system owners.
Side effects	None (documentation and design work).
Tests before change	Confirm the inventory entry matches the deployed configuration.
Tests after change	Inventory review signed off by the system owner.
Closure criteria	Inventory entry with owner, lifetime and target algorithm exists.

QSI-2026-06121 · P3 · 90 days short-term risk reduction

Add ECDSA usages at source code to the cryptographic inventory with an owner and a data-lifetime estimate; design for crypto-agility (algorithm identifiers, configurable providers).

Rationale	Migration cannot be planned without knowing where and why the algorithm is used.
Expected outcome	Every usage has an owner, a lifetime and a migration path.
Complexity	medium
Dependencies	Access to system owners.
Side effects	None (documentation and design work).
Tests before change	Confirm the inventory entry matches the deployed configuration.
Tests after change	Inventory review signed off by the system owner.
Closure criteria	Inventory entry with owner, lifetime and target algorithm exists.

QSI-2026-06123 · P3 · 90 days short-term risk reduction

Add Schnorr (BIP-340) usages at source code to the cryptographic inventory with an owner and a data-lifetime estimate; design for crypto-agility (algorithm identifiers, configurable providers).

Rationale	Migration cannot be planned without knowing where and why the algorithm is used.
Expected outcome	Every usage has an owner, a lifetime and a migration path.
Complexity	medium
Dependencies	Access to system owners.
Side effects	None (documentation and design work).
Tests before change	Confirm the inventory entry matches the deployed configuration.
Tests after change	Inventory review signed off by the system owner.

Closure criteria	Inventory entry with owner, lifetime and target algorithm exists.

QSI-2026-06122 · P3 · 90 days

short-term risk reduction

Add ECDSA usages at source code to the cryptographic inventory with an owner and a data-lifetime estimate; design for crypto-agility (algorithm identifiers, configurable providers).

Rationale	Migration cannot be planned without knowing where and why the algorithm is used.
Expected outcome	Every usage has an owner, a lifetime and a migration path.
Complexity	medium
Dependencies	Access to system owners.
Side effects	None (documentation and design work).
Tests before change	Confirm the inventory entry matches the deployed configuration.
Tests after change	Inventory review signed off by the system owner.
Closure criteria	Inventory entry with owner, lifetime and target algorithm exists.

QSI-2026-06124 · P3 · 90 days

short-term risk reduction

Add ECDH usages at source code to the cryptographic inventory with an owner and a data-lifetime estimate; design for crypto-agility (algorithm identifiers, configurable providers).

Rationale	Migration cannot be planned without knowing where and why the algorithm is used.
Expected outcome	Every usage has an owner, a lifetime and a migration path.
Complexity	medium
Dependencies	Access to system owners.
Side effects	None (documentation and design work).
Tests before change	Confirm the inventory entry matches the deployed configuration.
Tests after change	Inventory review signed off by the system owner.
Closure criteria	Inventory entry with owner, lifetime and target algorithm exists.

QSI-2026-06125 · P3 · 90 days

short-term risk reduction

Add Schnorr (BIP-340) usages at source code to the cryptographic inventory with an owner and a data-lifetime estimate; design for crypto-agility (algorithm identifiers, configurable providers).

Rationale	Migration cannot be planned without knowing where and why the algorithm is used.
Expected outcome	Every usage has an owner, a lifetime and a migration path.
Complexity	medium
Dependencies	Access to system owners.
Side effects	None (documentation and design work).
Tests before change	Confirm the inventory entry matches the deployed configuration.
Tests after change	Inventory review signed off by the system owner.

Closure criteria	Inventory entry with owner, lifetime and target algorithm exists.

QSI-2026-06126 · P3 · 90 days

short-term risk reduction

Add ECDSA usages at source code to the cryptographic inventory with an owner and a data-lifetime estimate; design for crypto-agility (algorithm identifiers, configurable providers).

Rationale	Migration cannot be planned without knowing where and why the algorithm is used.
Expected outcome	Every usage has an owner, a lifetime and a migration path.
Complexity	medium
Dependencies	Access to system owners.
Side effects	None (documentation and design work).
Tests before change	Confirm the inventory entry matches the deployed configuration.
Tests after change	Inventory review signed off by the system owner.
Closure criteria	Inventory entry with owner, lifetime and target algorithm exists.

QSI-2026-06120 · P4 · 90 days

root-cause fix

Plan migration of RSA to ML-DSA (FIPS 204) or SLH-DSA (FIPS 205), in a hybrid/ composite form where the ecosystem supports it. (or a hybrid scheme) once supported by your libraries, protocols and counterparties.

Rationale	Removes dependence on a quantum-vulnerable public-key primitive.
Expected outcome	The asset uses a NIST-standardized post-quantum or hybrid algorithm.
Complexity	high
Dependencies	Library/protocol support (e.g. OpenSSL ≥ 3.5, HSM firmware), interoperability with peers.
Side effects	Larger keys/signatures, possible performance and compatibility impact.
Tests before change	Interoperability tests in a staging environment.
Tests after change	Verify the negotiated/used algorithm on the production asset.
Closure criteria	Re-audit shows ML-DSA (FIPS 204) or SLH-DSA (FIPS 205), in a hybrid/ composite form where the ecosystem supports it. or a hybrid scheme in use.

QSI-2026-06121 · P4 · 90 days

root-cause fix

Plan migration of ECDSA to ML-DSA (FIPS 204) or SLH-DSA (FIPS 205), in a hybrid/ composite form where the ecosystem supports it (or a hybrid scheme) once supported by your libraries, protocols and counterparties.

Rationale	Removes dependence on a quantum-vulnerable public-key primitive.
Expected outcome	The asset uses a NIST-standardized post-quantum or hybrid algorithm.
Complexity	high
Dependencies	Library/protocol support (e.g. OpenSSL ≥ 3.5, HSM firmware), interoperability with peers.
Side effects	Larger keys/signatures, possible performance and compatibility impact.
Tests before change	Interoperability tests in a staging environment.
Tests after change	Verify the negotiated/used algorithm on the production asset.

Closure criteria	Re-audit shows ML-DSA (FIPS 204) or SLH-DSA (FIPS 205), in a hybrid/ composite form where the ecosystem supports it or a hybrid scheme in use.

QSI-2026-06123 · P4 · 90 days

root-cause fix

Plan migration of Schnorr (BIP-340) to (or a hybrid scheme) once supported by your libraries, protocols and counterparties.

Rationale	Removes dependence on a quantum-vulnerable public-key primitive.
Expected outcome	The asset uses a NIST-standardized post-quantum or hybrid algorithm.
Complexity	high
Dependencies	Library/protocol support (e.g. OpenSSL ≥ 3.5, HSM firmware), interoperability with peers.
Side effects	Larger keys/signatures, possible performance and compatibility impact.
Tests before change	Interoperability tests in a staging environment.
Tests after change	Verify the negotiated/used algorithm on the production asset.
Closure criteria	Re-audit shows or a hybrid scheme in use.

QSI-2026-06122 · P4 · 90 days

root-cause fix

Plan migration of ECDSA to ML-DSA (FIPS 204) or SLH-DSA (FIPS 205), in a hybrid/ composite form where the ecosystem supports it (or a hybrid scheme) once supported by your libraries, protocols and counterparties.

Rationale	Removes dependence on a quantum-vulnerable public-key primitive.
Expected outcome	The asset uses a NIST-standardized post-quantum or hybrid algorithm.
Complexity	high
Dependencies	Library/protocol support (e.g. OpenSSL ≥ 3.5, HSM firmware), interoperability with peers.
Side effects	Larger keys/signatures, possible performance and compatibility impact.
Tests before change	Interoperability tests in a staging environment.
Tests after change	Verify the negotiated/used algorithm on the production asset.
Closure criteria	Re-audit shows ML-DSA (FIPS 204) or SLH-DSA (FIPS 205), in a hybrid/ composite form where the ecosystem supports it or a hybrid scheme in use.

QSI-2026-06124 · P4 · 90 days

root-cause fix

Plan migration of ECDH to Hybrid key establishment with ML-KEM (FIPS 203), e.g. X25519MLKEM768 in TLS 1.3 or mlkem768x25519-sha256 in SSH (or a hybrid scheme) once supported by your libraries, protocols and counterparties.

Rationale	Removes dependence on a quantum-vulnerable public-key primitive.
Expected outcome	The asset uses a NIST-standardized post-quantum or hybrid algorithm.
Complexity	high
Dependencies	Library/protocol support (e.g. OpenSSL ≥ 3.5, HSM firmware), interoperability with peers.
Side effects	Larger keys/signatures, possible performance and compatibility impact.
Tests before change	Interoperability tests in a staging environment.
Tests after change	Verify the negotiated/used algorithm on the production asset.

Closure criteria	Re-audit shows Hybrid key establishment with ML-KEM (FIPS 203), e.g. X25519MLKEM768 in TLS 1.3 or mlkem768x25519-sha256 in SSH or a hybrid scheme in use.

QSI-2026-06125 · P4 · 90 days root-cause fix

Plan migration of Schnorr (BIP-340) to (or a hybrid scheme) once supported by your libraries, protocols and counterparties.

Rationale	Removes dependence on a quantum-vulnerable public-key primitive.
Expected outcome	The asset uses a NIST-standardized post-quantum or hybrid algorithm.
Complexity	high
Dependencies	Library/protocol support (e.g. OpenSSL ≥ 3.5, HSM firmware), interoperability with peers.
Side effects	Larger keys/signatures, possible performance and compatibility impact.
Tests before change	Interoperability tests in a staging environment.
Tests after change	Verify the negotiated/used algorithm on the production asset.
Closure criteria	Re-audit shows or a hybrid scheme in use.

QSI-2026-06126 · P4 · 90 days root-cause fix

Plan migration of ECDSA to ML-DSA (FIPS 204) or SLH-DSA (FIPS 205), in a hybrid/ composite form where the ecosystem supports it (or a hybrid scheme) once supported by your libraries, protocols and counterparties.

Rationale	Removes dependence on a quantum-vulnerable public-key primitive.
Expected outcome	The asset uses a NIST-standardized post-quantum or hybrid algorithm.
Complexity	high
Dependencies	Library/protocol support (e.g. OpenSSL ≥ 3.5, HSM firmware), interoperability with peers.
Side effects	Larger keys/signatures, possible performance and compatibility impact.
Tests before change	Interoperability tests in a staging environment.
Tests after change	Verify the negotiated/used algorithm on the production asset.
Closure criteria	Re-audit shows ML-DSA (FIPS 204) or SLH-DSA (FIPS 205), in a hybrid/ composite form where the ecosystem supports it or a hybrid scheme in use.

QSI-2026-06127 · P4 · 90 days root-cause fix

Update the development and build dependencies listed in the evidence.

Rationale	Eliminates the root cause described in the finding.
Expected outcome	The weakness is no longer present on re-test.
Complexity	low
Dependencies	Change window and owner approval.
Side effects	Tooling upgrades may change build output; compare release artefacts.
Tests before change	Reproduce the finding using the documented steps.
Tests after change	Re-run the retest procedure from the finding.
Closure criteria	

The dependency scan reports no advisories for tooling packages, or the remaining ones are recorded.

Priorities for 30, 60 and 90 days

30 days

Priority	ID	Action	Type
None			

60 days

Priority	ID	Action	Type
None			

90 days

Priority	ID	Action	Type
P3	QSI-2026-06120	Add RSA usages at file:components/libwally-core/upstream/SECURITY.md to the cryptographic inventory with an owner and a data-lifetime estimate; design for crypto-agility (algorithm identifiers, configurable providers).	short-term risk reduction
P3	QSI-2026-06121	Add ECDSA usages at source code to the cryptographic inventory with an owner and a data-lifetime estimate; design for crypto-agility (algorithm identifiers, configurable providers).	short-term risk reduction
P3	QSI-2026-06123	Add Schnorr (BIP-340) usages at source code to the cryptographic inventory with an owner and a data-lifetime estimate; design for crypto-agility (algorithm identifiers, configurable providers).	short-term risk reduction
P3	QSI-2026-06122	Add ECDSA usages at source code to the cryptographic inventory with an owner and a data-lifetime estimate; design for crypto-agility (algorithm identifiers, configurable providers).	short-term risk reduction
P3	QSI-2026-06124	Add ECDH usages at source code to the cryptographic inventory with an owner and a data-lifetime estimate; design for crypto-agility (algorithm identifiers, configurable providers).	short-term risk reduction
P3	QSI-2026-06125	Add Schnorr (BIP-340) usages at source code to the cryptographic inventory with an owner and a data-lifetime estimate; design for crypto-agility (algorithm identifiers, configurable providers).	short-term risk reduction
P3	QSI-2026-06126	Add ECDSA usages at source code to the cryptographic inventory with an owner and a data-lifetime estimate; design for crypto-agility (algorithm identifiers, configurable providers).	short-term risk reduction
P4	QSI-2026-06120	Plan migration of RSA to ML-DSA (FIPS 204) or SLH-DSA (FIPS 205), in a hybrid/composite form where the ecosystem supports it. (or a hybrid scheme) once supported by your libraries, protocols and counterparties.	root-cause fix
P4	QSI-2026-06121	Plan migration of ECDSA to ML-DSA (FIPS 204) or SLH-DSA (FIPS 205), in a hybrid/composite form where the ecosystem supports it (or a hybrid scheme) once supported by your libraries, protocols and counterparties.	root-cause fix
P4	QSI-2026-06123	Plan migration of Schnorr (BIP-340) to (or a hybrid scheme) once supported by your libraries, protocols and counterparties.	root-cause fix

P4	QSI-2026-06122	Plan migration of ECDSA to ML-DSA (FIPS 204) or SLH-DSA (FIPS 205), in a hybrid/composite form where the ecosystem supports it (or a hybrid scheme) once supported by your libraries, protocols and counterparties.	root-cause fix
P4	QSI-2026-06124	Plan migration of ECDH to Hybrid key establishment with ML-KEM (FIPS 203), e.g. X25519MLKEM768 in TLS 1.3 or mlkem768x25519-sha256 in SSH (or a hybrid scheme) once supported by your libraries, protocols and counterparties.	root-cause fix
P4	QSI-2026-06125	Plan migration of Schnorr (BIP-340) to (or a hybrid scheme) once supported by your libraries, protocols and counterparties.	root-cause fix
P4	QSI-2026-06126	Plan migration of ECDSA to ML-DSA (FIPS 204) or SLH-DSA (FIPS 205), in a hybrid/composite form where the ecosystem supports it (or a hybrid scheme) once supported by your libraries, protocols and counterparties.	root-cause fix
P4	QSI-2026-06127	Update the development and build dependencies listed in the evidence.	root-cause fix

Verification after remediation

ID	Retest procedure
QSI-2026-06120	Re-audit after migration.
QSI-2026-06121	Re-run the inventory after refactoring.
QSI-2026-06123	Re-run the inventory after refactoring.
QSI-2026-06122	Re-run the inventory after refactoring.
QSI-2026-06124	Re-run the inventory after refactoring.
QSI-2026-06125	Re-run the inventory after refactoring.
QSI-2026-06126	Re-run the inventory after refactoring.
QSI-2026-06127	Re-run the dependency scan after updating.

Tools used

Tool	Version
blockchain	1.0.0
certificates	1.1.0
chain_map	1.0.0
configs	1.1.0
dependencies	1.2.0
inventory_document	1.1.0
solidity	1.2.0
source_code	1.3.0
tls_code	1.1.0
tls_live	1.0.0
qsi-engine	1.2.0
OSV.dev API	v1

Sources and normative documents

Reviewed: 2026-10-13

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2. C. Chevignard, P.-A. Fouque, A. Schrottenloher — Reducing the Number of Qubits in Quantum Discrete Logarithms on Elliptic Curves (IACR ePrint 2026/280) — IACR ePrint. https://eprint.iacr.org/2026/280 (revision of a EUROCRYPT 2026 paper: 3.12n + o(n) qubits; for n = 256, 1193 qubits and 22 runs of 2^38.98 Toffoli gates)
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10. FIPS 204 — Module-Lattice-Based Digital Signature Standard (ML-DSA) — NIST. https://csrc.nist.gov/pubs/fips/204/final (final (August 2024))
11. FIPS 205 — Stateless Hash-Based Digital Signature Standard (SLH-DSA) — NIST. https://csrc.nist.gov/pubs/fips/205/final (final (August 2024))
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Appendices

A. Cryptographic inventory

Algorithm	Parameter	Quantum status	Classical status	Count	Locations	Examples
RIPEMD-160	RIPEMD-160	weak classically	legacy	2	2	components/libwally-core/upstream/src/ccan/ccan/crypto/ripemd160/ripemd160.c; components/libwally-core/upstream/src/ccan/ccan/crypto/ripemd160/ripemd160.h
HMAC		reduced margin (Grover), approved	acceptable	9	9	components/libwally-core/upstream/include/wally.hpp:146; components/libwally-core/upstream/include/wally_bip32.h:170; components/libwally-core/upstream/src/aes.c:273
PBKDF2		reduced margin (Grover), approved	acceptable	1	1	components/libwally-core/upstream/src/pbkdf2.c
SHA-2	SHA-512	reduced margin (Grover), approved	acceptable	2	2	components/libwally-core/upstream/src/ccan/ccan/crypto/sha512/sha512.c; components/libwally-core/upstream/src/ccan/ccan/crypto/sha512/sha512.h
SHAKE	SHAKE256	reduced margin (Grover), approved	acceptable	2	2	main/utils/shake256.c; main/utils/shake256.h
scrypt		reduced margin (Grover), approved	acceptable	5	5	components/libwally-core/upstream/src/scrypt.c; components/libwally-core/upstream/src/scrypt/crypto_scrypt.c; components/libwally-core/upstream/src/scrypt/crypto_scrypt_smix.c
SHA-2		Grover: acceptable to NIST, below ECCG/ ANSSI guidance	acceptable	2	2	components/esp32_bc-ur/src/sha2.c; components/esp32_bc-ur/src/sha2.h
SHA-2	SHA-256	Grover: acceptable to NIST, below ECCG/ ANSSI guidance	acceptable	4	4	components/libwally-core/upstream/src/ccan/ccan/crypto/sha256/sha256.c; components/libwally-core/upstream/src/ccan/ccan/crypto/sha256/sha256.h; components/libwally-core/upstream/src/ccan/ccan/crypto/sha256/sha256_sse4.c

ECDH		vulnerable (Shor)	acceptable	3	3	components/libwally-core/upstream/src/ecdh.c; components/libwally-core/upstream/src/secp256k1/include/secp256k1_ecdh.h; components/libwally-core/upstream/src/secp256k1/src/bench.c:24
ECDSA	secp256k1	vulnerable (Shor)	acceptable	36	36	components/libwally-core/upstream/setup.py:58; components/libwally-core/upstream/src/amalgamation/combined.c:34; components/libwally-core/upstream/src/internal.h:33
ECDSA		vulnerable (Shor)	acceptable	16	16	components/libwally-core/upstream/src/anti_exfil.c:72; components/libwally-core/upstream/src/secp256k1/contrib/lax_der_parsing.c:11; components/libwally-core/upstream/src/secp256k1/contrib/lax_der_parsing.h:88
RSA	4096-bit	vulnerable (Shor)	acceptable	2	2	components/libwally-core/upstream/SECURITY.md (line 52)
RSA		vulnerable (Shor)	acceptable	2	2	main/rsa.c; main/rsa.h
Schnorr (BIP-340)		vulnerable (Shor)	acceptable	7	7	components/libwally-core/upstream/src/secp256k1/include/secp256k1_schnorrsig.h:82; components/libwally-core/upstream/src/secp256k1/include/secp256k1_schnorrsig_halfagg.h:43; components/libwally-core/upstream/src/secp256k1/src/modules/musig/session_impl.h:621

Long lists are grouped here; the full per-file list is in the JSON report, signed with the same key.

B. JSON

qsi.report/1 · SHA-256 7942ddb55815a12070642368e6e3ea6fc9c02c06f4bad7fe1df427ff83cae3e3

C. Input files

File	Size	SHA-256
components/assets/assets_snapshot.c	2103	ec1f77a3fc4474a97c955a063a815513508ae
components/assets/assets_snapshot.h	514	a680e722bb23d471e95bbc65a990620dd383a
components/assets/gen_assets.py	2715	9d8beedc80645bb85c3892b0d2532e4bf3901
components/autogenlang/gen_locale.py	3762	5d86c480623ba04ca88a4eb227fe52ac22acc
components/autogenlang/jlocale.c	1083	a2867e2942a695dd8f91dab15cf6638339bd9
components/autogenlang/jlocale.h	703	237a507c4e96730d74f1966ff903c984bcaf6

components/esp32-quirc/lib/decode.c	22800	eaae29c6cdc2b68e996dcc8fe1e17669a4c9b
components/esp32-quirc/lib/identify.c	33069	932e3f59f2a53449a6f9a96711f6b235f3309
components/esp32-quirc/lib/quirc.c	2643	098d6fcff90f0f1b6e567bcd0106a9f2e3b28
components/esp32-quirc/lib/quirc.h	5789	4ae430f11eae91c19e497a5c6a33e8ae3d639
components/esp32-quirc/lib/quirc_internal.h	3256	8f91d08224445af8f59b22c3ee7cbd8125b66
components/esp32-quirc/lib/version_db.c	9934	1313ca98abb0342f1879cbc7cf2a2f7a6ec6c
components/esp32-quirc/openmv/collections.c	1662	832f54697073cc3476e4eaddc73362bf23780
components/esp32-quirc/openmv/collections.h	965	bb8700a2f2be2c8c4694896df16db1263df4c
components/esp32-quirc/openmv/fmath.h	1619	162de4e6e80de93f2ed28eafbfa16d90052c1
components/esp32-rotary-encoder/include/rotary_encoder.h	7247	b3ac17bb6ec08b1170f5806c55db7789b6271
components/esp32-rotary-encoder/rotary_encoder.c	11045	1de7b32b77f37f3023f67c4161873b8f1632c
components/esp32_bc-ur/CLA-signed/ CLA.gorazdko.41F0EA1699A74C1E2FA41B538CF96BC3FF9DBBCE.asc	4412	7d33b77a5eb135e9b5a975edc67491f236f4a
components/esp32_bc-ur/CLA-signed/ CLA.wolfmcnally.943652EE38441760C3DC35364B6C2FCF894780AE.asc	4028	d304435a759e2cb6e2fa2c7e16d84c4b2b15b
components/esp32_bc-ur/src/bc-ur.hpp	528	40df2bfc8295525203a349c8ae60e8e965ad3
components/esp32_bc-ur/src/bytewords.cpp	7264	78eb4d49e8435db5aacf352fa080ef4d388cd
components/esp32_bc-ur/src/bytewords.hpp	528	7589b1785de98f2f845dccc06d3dbf13f8c8f
components/esp32_bc-ur/src/cbor-lite.hpp	20610	896dec208c574224420b1415131e0b3916e83
components/esp32_bc-ur/src/cdecoder.h	998	f95f23b5c1c6203f785a3a1fd2c9a93d2345f
components/esp32_bc-ur/src/cencoder.h	1017	e942fff6390e7a29db59dbf4cd1c73204db71
components/esp32_bc-ur/src/crc32.c	975	42a40df27042ba588a75c9dbb81fddb13ccdf
components/esp32_bc-ur/src/crc32.h	556	dd772af1a4c3424e02ea8a78a5b1e47e8c4e0
components/esp32_bc-ur/src/fountain-decoder.cpp	6560	1db9c5920b4f3451e32aa29a0d35000fcc187
components/esp32_bc-ur/src/fountain-decoder.hpp	3319	3feaaf5a28145c3812875f30f843067486e59
components/esp32_bc-ur/src/fountain-encoder.cpp	6696	23d95af7d1e0bfad0e9ba2abd45ae14c03e72
components/esp32_bc-ur/src/fountain-encoder.hpp	2525	1928f66b4f66b2c7c0b0a0bb7944430c3a0b6
components/esp32_bc-ur/src/fountain-utils.cpp	2122	4b2bb5060c09d049e7f702157d649e0968fa5
components/esp32_bc-ur/src/fountain-utils.hpp	1308	a389b161d33037fa6d1f3817508c1c6c12270
components/esp32_bc-ur/src/memzero.c	2371	e079c96d106e8fb94792a84633d491b178151
components/esp32_bc-ur/src/memzero.h	146	b97b8177b7e8a5145f1924a1195bb4228305
components/esp32_bc-ur/src/psram-allocator.hpp	1074	e0b889435117a283decd54954a229b23c98c3
components/esp32_bc-ur/src/random-sampler.cpp	2005	9db3b85bb093ab9a634318b6e7e1784fc3a55
components/esp32_bc-ur/src/random-sampler.hpp	778	ed32061f179179b4f220affa3629b2ce9f556

components/esp32_bc-ur/src/sha2.c	27586	f2d198651782b1fd9ce32308551955486b8a3
components/esp32_bc-ur/src/sha2.h	4018	33c81cba2aca364b74a0205e1c294795046f2
components/esp32_bc-ur/src/ur-decoder.cpp	6206	561364d15d528855dff672332700bd4ff01db
components/esp32_bc-ur/src/ur-decoder.hpp	2427	731ca04a557bcdbccbbeab1bedbdc3d830f3a
components/esp32_bc-ur/src/ur-encoder.cpp	4142	719edb19a5cfe0aab44822447be5817782432
components/esp32_bc-ur/src/ur-encoder.hpp	1698	70fbdfb8edeeaafc59319880a01aa0e63d5f8
components/esp32_bc-ur/src/ur.cpp	536	a4a9ef9c96c9619d4f3fb055cbcfd44fc6a3b
components/esp32_bc-ur/src/ur.hpp	629	aa99021077e929b2ddc37190e7bd65db0fd7b
components/esp32_bc-ur/src/utils.cpp	1980	e9eb8c577fd22f7463c86856f61539142bb26
components/esp32_bc-ur/src/utils.hpp	2201	7f6484296e698ba343d1ec6f1e7f93809ab00
components/esp32_bc-ur/src/xoshiro256.cpp	2027	37c9183a02c28859c92422f0c69de314ccbae
components/esp32_bc-ur/src/xoshiro256.hpp	530	d72cd4a37d04a8553196a120d26132e1e57a6
components/esp32_bsdiff/bsdiff.c	9697	e03e27e1f289c7746bf3734245a77fbcc6bb6
components/esp32_bsdiff/bsdiff.h	1738	0c62584472d7b1e97ee59221614b5270512fe
components/esp32_bsdiff/bspatch.c	6267	49727e4bcd81767180365c91f252bcb89b214
components/esp32_bsdiff/bspatch.h	1951	9f6d00b6b892e168f95c935eb4f98a74914c7
components/esp32_deflate/deflate.c	7375	83f4ce0e28fda75cdd5ba7ef3cdac5ba4f319
components/esp32_deflate/deflate.h	3881	51e86c8beb6991d8e7aeaa81eabbc11a2df2b
components/google-otpauth-migrate/google-otpauth-migration.pb.c	391	d75daa9efdf81db881563a904ca354d1c05c3
components/google-otpauth-migrate/google-otpauth-migration.pb.h	5866	0a37cbc704ef223415355e579f4077be77b2c
components/google-otpauth-migrate/google-otpauth-migration.proto	746	0661efbbeaf3b3d99ac84372a75ae45e0c60e
components/google-otpauth-migrate/pb.h	44863	8da6125d3b5e650781db012a97d3dd7d25868
components/google-otpauth-migrate/pb_common.c	12141	8d2ec28baaaf2b7a5e90e4cb2fa9700d21cef
components/google-otpauth-migrate/pb_common.h	1677	6495a691aca68d6973f2274b5dd54b74fbb57
components/google-otpauth-migrate/pb_decode.c	55680	3c3394300a93a152ac6280786a2d98ed57e00
components/google-otpauth-migrate/pb_decode.h	7870	1747746e5961de5789bcf0795588da0790cd1
components/libwally-core/config.h	5305	ba770c0e09abb8fa74523560cc7e6b24fe6a7
components/libwally-core/upstream/SECURITY.md	5780	3ab230590b464090a9bd2d92a77fe84b7f651
components/libwally-core/upstream/contrib/requirements.txt	76	40039c059b565df34460f06ccbc2e54a86f75
components/libwally-core/upstream/docs/source/conf.py	9400	bc6dfa797847a290d783f88a90731a5b2e28c
components/libwally-core/upstream/include/wally.hpp	173558	29fe67c31ac2364b9bcbddf331b1a2547c348
components/libwally-core/upstream/include/wally_address.h	14902	7f7c7746344d85a5d4c4dd294eacb28d93f80
components/libwally-core/upstream/include/wally_anti_exfil.h	4298	3b5dc9737f3c2921ef1ef8f8b14254b705fe2

components/libwally-core/upstream/include/wally_bip32.h	18820	810cb696a773ebfc473140d4500466f4755b0
components/libwally-core/upstream/include/wally_bip38.h	4113	21b2a312a50b325c239960e44dbbfc71ed8c0
components/libwally-core/upstream/include/wally_bip39.h	4675	37d73b2b2c5477c56b5f226f8e6a67595f813
components/libwally-core/upstream/include/wally_bip85.h	2147	e905f637ff408e6e406f9b4ee292be6e4592
components/libwally-core/upstream/include/wally_coinselection.h	1952	7e8ffb7f18e814876726e5dc7338b745b6924
components/libwally-core/upstream/include/wally_core.h	14337	3022b6eee2211127b4b29afc07e9efa6a3b43
components/libwally-core/upstream/include/wally_crypto.h	36912	2181a542b32bbf426de0259360751fa6e821c
components/libwally-core/upstream/include/wally_descriptor.h	18149	9651e5cd915347dc9033a9a3e8f69698f762b
components/libwally-core/upstream/include/wally_elements.h	30713	12520fe83d0d7a6985a8f89f3483465920a56
components/libwally-core/upstream/include/wally_map.h	23352	49e8504df27c0a3530f125fb11bf8b6476a9e
components/libwally-core/upstream/include/wally_psbt.h	92379	5f22a3900c2be3fba63b3f0e48fe36509c7db
components/libwally-core/upstream/include/wally_psbt_members.h	28550	02f5d3b6038d0ddc79830686dcfe05f92f5c0
components/libwally-core/upstream/include/wally_script.h	24734	f9714d64df46d9282d0af7295f3b5efad8255
components/libwally-core/upstream/include/wally_symmetric.h	1403	c592f068170a611a0b4f45610f8af1901f25f
components/libwally-core/upstream/include/wally_transaction.h	55083	02f09466ce2e7bc8f326f762efde701d02c84
components/libwally-core/upstream/include/wally_transaction_members.h	14257	71199aa909d73759bcb9328885072d810bfab
components/libwally-core/upstream/pyproject.toml	91	81562b11a9a84cb5d47f66a78e7338546fddb
components/libwally-core/upstream/setup.py	7382	016931c29406077b6d11f67500ba0bc5cc27e
components/libwally-core/upstream/src/address.c	8162	f7faee93c6eec3fa6311ef59e30aee3aa5c38
components/libwally-core/upstream/src/aes.c	13043	5a9d9d02276f306f18a1afec725d00544d43c
components/libwally-core/upstream/src/amalgamation/combined.c	3370	ea2c2393b4f2f218bed700c4d880fdff8f649
components/libwally-core/upstream/src/amalgamation/windows_config/config.h	518	1b14c47566007e84ccf825e180841132f485f
components/libwally-core/upstream/src/anti_exfil.c	4182	9a82b2e8a8fb25df7d59ee0108552079c29c7
components/libwally-core/upstream/src/base58.h	329	19d52a6c48e467662faf41cbe7bcaf86850dc
components/libwally-core/upstream/src/base_58.c	10621	2765b178cc94418c37ac520fbbcd7082fb817
components/libwally-core/upstream/src/base_64.c	2352	fdd2a5e08206baee701ede543348dcf13f623
components/libwally-core/upstream/src/bech32.c	11107	fbef184ce2e675180e1d4b08294fde0d22aac
components/libwally-core/upstream/src/bip32.c	42775	ebe4939b794fa8d447ef1054296842c5e1fa3
components/libwally-core/upstream/src/bip32_int.h	1789	00614706452155f6adcc4699a3bee9537ccb7
components/libwally-core/upstream/src/bip38.c	13750	7ded4248d2176e6f18ef02d556bc138872d19
components/libwally-core/upstream/src/bip39.c	7536	0b4ad66758f2f02a9fa5dbaa1daad4094deb8
components/libwally-core/upstream/src/bip85.c	4885	4eeff959873565157d2c8696536bdbb3db402

components/libwally-core/upstream/src/blech32.c	11446	6f115f7716d84eb277e942fcb281b55a229b0
components/libwally-core/upstream/src/ccan/ccan/base64/base64.c	7742	2660b9eeb6c66e0a8187f8b946471f1d9463b
components/libwally-core/upstream/src/ccan/ccan/base64/base64.h	8587	75e1376547570c23b5cb51a58b7eb06e00e22
components/libwally-core/upstream/src/ccan/ccan/build_assert/build_assert.h	1228	eeebbbc75d152ab145fee9ea5fb3e9148220
components/libwally-core/upstream/src/ccan/ccan/compiler/compiler.h	6208	265ec2faa538935602f4d9527e027aff5d220
components/libwally-core/upstream/src/ccan/ccan/crypto/ripemd160/ripemd160.c	14205	4d231f50f83073f1a4c0a79f9c3e755e9350b
components/libwally-core/upstream/src/ccan/ccan/crypto/ripemd160/ripemd160.h	4229	7f7d917d859e20b40c95e94b9a4862eda420c
components/libwally-core/upstream/src/ccan/ccan/crypto/sha256/sha256.c	12527	286fc60aeaef4c3f23e61f5c43068b39fbfa1
components/libwally-core/upstream/src/ccan/ccan/crypto/sha256/sha256.h	4392	d68d0b50a1695e3762dc6a18f3e15de2c270c
components/libwally-core/upstream/src/ccan/ccan/crypto/sha256/sha256_sse4.c	26723	97974a76fb8ae60ea7b3f10e1f885631b38bc
components/libwally-core/upstream/src/ccan/ccan/crypto/sha512/sha512.c	12350	acfe1bc068339cee74a59ec420d11a1161fa3
components/libwally-core/upstream/src/ccan/ccan/crypto/sha512/sha512.h	3799	2e013c21c38bdcecd5407f21a7cc7edb55d62
components/libwally-core/upstream/src/ccan/ccan/endian/endian.h	9168	079dc923af35babc0a19d48920453eac806aa
components/libwally-core/upstream/src/ccan/ccan/str/hex/hex.c	1240	f19e45cfcd6f9e1a57135d0ab0f7664467a50
components/libwally-core/upstream/src/ccan/ccan/str/hex/hex.h	1907	bb26783b82e632be5a7bc2af185655119124
components/libwally-core/upstream/src/ccan/ccan/tap/tap.c	9251	44ac24b231f4a27d49e3a14bd02539491174
components/libwally-core/upstream/src/ccan/ccan/tap/tap.h	8136	0349b6db4d1699703fae8ad17f0d8cd3c585a
components/libwally-core/upstream/src/ccan_config.h	1856	fbfef65cb854aff875e982a0f3f49913a7e13
components/libwally-core/upstream/src/coins.c	7481	32a65ff8892db2fa25f1ea43e4c5e4ae5dd2c
components/libwally-core/upstream/src/cpufeatures/cpu-features.c	40717	60b4e0194012595a58c88fcd4089bc72c4752
components/libwally-core/upstream/src/cpufeatures/cpu-features.h	11746	4e85e43209f0ba184a4bc5d3353d99695d4b7
components/libwally-core/upstream/src/ctaes/bench.c	4977	237677992b69e7dcc6d4fab66f74e8afa7c99
components/libwally-core/upstream/src/ctaes/ctaes.c	18015	2f8a63e6a9a00da3fcc9c38d53dfbbb650376
components/libwally-core/upstream/src/ctaes/ctaes.h	1541	bf837cfa914ac30b11cea64d5267065614686
components/libwally-core/upstream/src/ctest/psbts.h	96356	03ef1939fb98779fe78fd0619ac9af38b4b9f
components/libwally-core/upstream/src/data/wordlists/chinese_simplified.c	66952	0b5757eb83ab683d573f2f33a8f6beb55206f
components/libwally-core/upstream/src/data/wordlists/chinese_traditional.c	66952	e3c7f2043ed23fcc05a2d56c4a0fde20fb8d7

components/libwally-core/upstream/src/data/wordlists/english.c	90123	6281a94bbb2f078bf81c5e896ba256bfa647e
components/libwally-core/upstream/src/data/wordlists/french.c	108796	456ccc99b4e1deb3c32be31cd1da15fc2c24f
components/libwally-core/upstream/src/data/wordlists/italian.c	104983	a62efd66b93dba13bd6595c75223c16fba1c6
components/libwally-core/upstream/src/data/wordlists/japanese.c	157532	6e05670b15966f5ed2f9dda1eb65176ef0db0
components/libwally-core/upstream/src/data/wordlists/spanish.c	94627	2868a35904d7a59825854b1c7aa74b1ceb482
components/libwally-core/upstream/src/descriptor.c	135286	67db7c1b957cef26b1b8c3c75804c8e0e0d2c
components/libwally-core/upstream/src/ecdh.c	909	8ee530f4d9ae4d9bf59d56be587898902ef61
components/libwally-core/upstream/src/elements.c	45442	f2619a99c89c18b470b3d309f004bfa999948
components/libwally-core/upstream/src/hex_.c	1814	b2d657a4efdea61b6fc6499384a54e084ece2
components/libwally-core/upstream/src/hmac.c	1029	ec26247e22f48069c0b590cab1df5c9eb6405
components/libwally-core/upstream/src/hmac.h	945	272f5733e09fcd7bbed99e7090d6dfa45f280
components/libwally-core/upstream/src/internal.c	18775	fc35a46528e06386282ca70d31fde07b9aaee
components/libwally-core/upstream/src/internal.h	7781	f21377ed9eba49d06a5664dc4c28437e62d1b
components/libwally-core/upstream/src/map.c	32790	d51f09a338d5453dacd2586c6704008c06fa3
components/libwally-core/upstream/src/mnemonic.c	3004	8a1bb43751a5bfd8702a970eaa6707af12c97
components/libwally-core/upstream/src/mnemonic.h	975	74f5df50a738b55423f5920b0d3d4441c1cea
components/libwally-core/upstream/src/pbkdf2.c	889	105f023951c03024c125a18b0f5862fe766f5
components/libwally-core/upstream/src/psbt.c	243364	e76cac2111ddcdbd1abe201be5f2411cb4111
components/libwally-core/upstream/src/psbt_io.h	13779	851894fd3136feb0c911152c290491b5a858c
components/libwally-core/upstream/src/pullpush.c	7444	777d7edd1d1b649a2dcf32a7db0da3eab0254
components/libwally-core/upstream/src/pullpush.h	4514	b1d4ce8e2abed2de4edb1917f4822517e235c
components/libwally-core/upstream/src/pyexample/anti-exfil.py	660	1a667aaadb838d6e397bb99d1bf6b1cb5a463
components/libwally-core/upstream/src/pyexample/liquid/receive-send.py	22276	bc90191a9307600954861a8368f2c4b5a3599
components/libwally-core/upstream/src/script.c	46749	fb0fe6ddcd942103db5b0d2dc44259d154aea
components/libwally-core/upstream/src/script.h	818	1617a0849f8db48baa659756a8df6fd8cda1a
components/libwally-core/upstream/src/script_int.h	3734	bfb7c554290734d0fd4204ebcbad2496f46e4
components/libwally-core/upstream/src/scrypt/crypto_scrypt.c	7396	033c20dbf6d9353f01f45cf14752afea67811
components/libwally-core/upstream/src/scrypt/crypto_scrypt_smix.c	6241	87c79624c9a5b5d1375dd332734f14758fb87
components/libwally-core/upstream/src/scrypt/crypto_scrypt_smix_neon.c	9554	a92a7abf7fcd2cd406acd8a3fa37a16fddf05
components/libwally-core/upstream/src/scrypt/crypto_scrypt_smix_sse2.c	7101	5a0b698f710cdc38e951d574804fde85b74e6
components/libwally-core/upstream/src/scrypt.c	2523	7912d0bfe719e149ed99ed1dc097e3087e0a9

components/libwally-core/upstream/src/secp256k1/contrib/lax_der_parsing.c	3595	4da2ecdc0a3b9dbb9029d188bb0243764a3c2
components/libwally-core/upstream/src/secp256k1/contrib/lax_der_parsing.h	4230	7f3e8ba009f19a5f2759c9065af02cc8e6235
components/libwally-core/upstream/src/secp256k1/contrib/lax_der_privatekey_parsing.c	5112	48352c75520eecd141936c0011601925510cc
components/libwally-core/upstream/src/secp256k1/contrib/lax_der_privatekey_parsing.h	4016	3420812d623c0f2f86debe0535f98eab9b929
components/libwally-core/upstream/src/secp256k1/include/secp256k1.h	44268	e1717d8943d4a28ad0c7dc6c0b19d0afbc404
components/libwally-core/upstream/src/secp256k1/include/secp256k1_bppp.h	2761	bbd36b1d30ba8d181a84a234a29b91e1af6a6
components/libwally-core/upstream/src/secp256k1/include/secp256k1_ecdh.h	2560	ad314d14cb79a79d4113862f21e0deccf3796
components/libwally-core/upstream/src/secp256k1/include/secp256k1_ecdsa_adaptor.h	7476	5bb2f1c0e958c2380ff3dcc15e5f70d5ecf24
components/libwally-core/upstream/src/secp256k1/include/secp256k1_ecdsa_s2c.h	12237	ffcb82928ed6e09cd0e95e15cb3c67d95b191
components/libwally-core/upstream/src/secp256k1/include/secp256k1_ellswift.h	9148	620db468b1c13968946e9f404ceb59a3f11c8
components/libwally-core/upstream/src/secp256k1/include/secp256k1_extrakeys.h	10870	7c3468c9fd192ad152c5cf042cb44938961d7
components/libwally-core/upstream/src/secp256k1/include/secp256k1_generator.h	10816	1c08ef2a929eb52bff724c2d3d6e8d5a35538
components/libwally-core/upstream/src/secp256k1/include/secp256k1_musig.h	32175	e6e2595005881a64bce69a05cba3e2715e56a
components/libwally-core/upstream/src/secp256k1/include/secp256k1_preallocated.h	5966	b7283d4cc9c13bb2572dc92e69cea76f32704
components/libwally-core/upstream/src/secp256k1/include/secp256k1_rangeproof.h	11792	32ac0541545b6d8b562cd7e41043807fd06ae
components/libwally-core/upstream/src/secp256k1/include/secp256k1_recovery.h	5250	e00678a1108acb36c6631a1e290cf33813b0e
components/libwally-core/upstream/src/secp256k1/include/secp256k1_schnorrsig.h	8177	8185d6c6ea3234fff05f38fad4b1b6f48f057
components/libwally-core/upstream/src/secp256k1/include/secp256k1_schnorrsig_halfagg.h	4908	89669a3ad2c904072c1177e7b783996af5d72
components/libwally-core/upstream/src/secp256k1/include/secp256k1_surjectionproof.h	12407	f6b245fe8234309715225fb6a3fdf3a4d20a4
components/libwally-core/upstream/src/secp256k1/include/secp256k1_whitelist.h	6241	2e060720e5f8bdef9b99021a3c1d45a610219
components/libwally-core/upstream/src/secp256k1/src/assumptions.h	4667	cc875f356cc8f7f272def71f7d7261e44cc27
components/libwally-core/upstream/src/secp256k1/src/bench.c	10190	832edacf541809e3cbc59a4e698d5e33a9388

components/libwally-core/upstream/src/secp256k1/src/bench.h	4777	12fafa36167bb38d28698536cdb34bdbcf8c9
components/libwally-core/upstream/src/secp256k1/src/bench_bppp.c	1102	7c17a6b0fada57db18cc68b1ec183f1e63469
components/libwally-core/upstream/src/secp256k1/src/bench_ecmult.c	15684	7daaf2e676b42f08a509b846139e90f9f04da
components/libwally-core/upstream/src/secp256k1/src/bench_generator.c	2001	6396798ee1f4a018d4ebd9e962b6c289f4a19
components/libwally-core/upstream/src/secp256k1/src/bench_internal.c	19230	6fb683c1d9dd79261e9a519c033ee6934687f
components/libwally-core/upstream/src/secp256k1/src/bench_rangeproof.c	2349	6c24fa0c6c12242da3b2629742e0baab65b24
components/libwally-core/upstream/src/secp256k1/src/bench_whitelist.c	4089	8f77ebf85792d8270790e1c0689d24896c2da
components/libwally-core/upstream/src/secp256k1/src/checkmem.h	5355	f367089bffa53b8998047de2f718f75d627da
components/libwally-core/upstream/src/secp256k1/src/eccommit.h	2198	3b04b7fdae817984c26ce995fad4f3f6d72d5
components/libwally-core/upstream/src/secp256k1/src/eccommit_impl.h	3165	6c71a9cba341b7782302741603db3ae2179b6
components/libwally-core/upstream/src/secp256k1/src/ecdsa.h	1164	f97753416b0971133ad9d8acf9af02259931c
components/libwally-core/upstream/src/secp256k1/src/ecdsa_impl.h	10562	d396c1c758db5dffdc3044f00895bc1206f79
components/libwally-core/upstream/src/secp256k1/src/eckey.h	1449	cf511599285faecd27ace6191eca3945d85bc
components/libwally-core/upstream/src/secp256k1/src/eckey_impl.h	3439	4d327c593520c3f60f7b86d407df0870c26cf
components/libwally-core/upstream/src/secp256k1/src/ecmult.h	2768	1e87f35efb7e7e5b6ea1d6ec22bbaeab33ccd
components/libwally-core/upstream/src/secp256k1/src/ecmult_compute_table.h	1073	5a8ce69b18d777c3ad608ef173fd6a47c75b9
components/libwally-core/upstream/src/secp256k1/src/ecmult_compute_table_impl.h	1916	6082339f616cd5a628300df24e66353c1374a
components/libwally-core/upstream/src/secp256k1/src/ecmult_const.h	1354	4705e7baabcbf8fd1853f8cef45a7f6a3075b
components/libwally-core/upstream/src/secp256k1/src/ecmult_const_impl.h	19584	07238bcf17c3292bb8198f7f37baf068bdc9a
components/libwally-core/upstream/src/secp256k1/src/ecmult_gen.h	5458	38e1457b6e8ba8f488e5a0e19fb625df97603
components/libwally-core/upstream/src/secp256k1/src/ecmult_gen_compute_table.h	672	51eb54b0cb7b49b25cf751efb2799322d73c9
components/libwally-core/upstream/src/secp256k1/src/ecmult_gen_compute_table_impl.h	4705	9ae524743bb5871bff7e41e48b61218336b19
components/libwally-core/upstream/src/secp256k1/src/ecmult_gen_impl.h	16504	568bb11a3ecbbf116f13009b9e6de78c23fce
components/libwally-core/upstream/src/secp256k1/src/ecmult_impl.h	34325	ce70d26cb987c9bb2a2477389459437fc2831
components/libwally-core/upstream/src/secp256k1/src/field.h	15119	a031e8655f38464b47aedec8afd73b3320da5
components/libwally-core/upstream/src/secp256k1/src/field_10x26.h	2338	ba9a43b62e7e70e6e18af2787bd01dfc3a57f
components/libwally-core/upstream/src/secp256k1/src/field_10x26_impl.h	52106	6f209a533fd96e73f4c92b101bd3ca46a2f75
components/libwally-core/upstream/src/secp256k1/src/field_5x52.h	2380	cbaa91c253ba84d99efcd334c7a39906c164c

components/libwally-core/upstream/src/secp256k1/src/field_5x52_impl.h	19360	8104a1b630635c9cc89628142c9fa7afc6f31
components/libwally-core/upstream/src/secp256k1/src/field_5x52_int128_impl.h	11341	ee6d585731507c51dc8acd819b66556cf0cd8
components/libwally-core/upstream/src/secp256k1/src/field_impl.h	14052	33cb488ee009334a7a2ac52a7eabe39795bd1
components/libwally-core/upstream/src/secp256k1/src/group.h	12243	d5443ba9dc805fb378d6dcdab02bd616ab344
components/libwally-core/upstream/src/secp256k1/src/group_impl.h	35125	663da90fb18b6fc8dace045ceb28ea8fae6e1
components/libwally-core/upstream/src/secp256k1/src/hash.h	2625	a7f39f2ed170d70841eff7d43d9c480dfda6c
components/libwally-core/upstream/src/secp256k1/src/hash_impl.h	15448	70247571d95e3c8824d6ca2e90956f6941c19
components/libwally-core/upstream/src/secp256k1/src/hsort.h	1685	220cb669835d34c252598ecff860c93edda33
components/libwally-core/upstream/src/secp256k1/src/hsort_impl.h	4798	2861489c01b927b27d99efe173799a8124347
components/libwally-core/upstream/src/secp256k1/src/int128.h	3622	89ac5d91d64f857038d063dd70eec4b29b4f3
components/libwally-core/upstream/src/secp256k1/src/int128_impl.h	376	5683c02dfc495639da259d80624711c1c449b
components/libwally-core/upstream/src/secp256k1/src/int128_native.h	581	b4c52c7ff091083f5eca9879b98641c7a3835
components/libwally-core/upstream/src/secp256k1/src/int128_native_impl.h	2881	ecaafefadbde3db034d4f44678096833d2d11
components/libwally-core/upstream/src/secp256k1/src/int128_struct.h	229	0733b3a65719a1e9a8fb3564c71a38f2f31ca
components/libwally-core/upstream/src/secp256k1/src/int128_struct_impl.h	7884	0b4e2200327f5357078ba43b0e0514c58c2b1
components/libwally-core/upstream/src/secp256k1/src/modinv32.h	1865	774887ccd462b145434d1d6b9ae9455aa3618
components/libwally-core/upstream/src/secp256k1/src/modinv32_impl.h	33492	befe02af7099a871677c7b05a81302dff5a4c
components/libwally-core/upstream/src/secp256k1/src/modinv64.h	1969	b71cf177d768beac28fcb4f0ef545aafaad16
components/libwally-core/upstream/src/secp256k1/src/modinv64_impl.h	37653	1254dc7ea941fc4d4ef0a85eed80db58d9e26
components/libwally-core/upstream/src/secp256k1/src/modules/bppp/bppp_norm_product_impl.h	20378	13b376aadaf6ad496b82de06e57ce781c8dd5
components/libwally-core/upstream/src/secp256k1/src/modules/bppp/bppp_transcript_impl.h	1552	bc8b8afcf3e985bdb355e715c2f56f109e2fb
components/libwally-core/upstream/src/secp256k1/src/modules/bppp/bppp_util.h	2794	fb22fd546df224866adc6c756f76170b4aa25
components/libwally-core/upstream/src/secp256k1/src/modules/bppp/main.h	403	20d3c6914baf5513f17d02787e51a97ecaa16
components/libwally-core/upstream/src/secp256k1/src/modules/bppp/main_impl.h	3583	938885438c1a41562294d15e068e9a7dac589
components/libwally-core/upstream/src/secp256k1/src/modules/ecdh/bench_impl.h	1761	3929ead352b3b6425f0fdf94a1b2268d124cc
components/libwally-core/upstream/src/secp256k1/src/modules/ecdh/main_impl.h	2929	3cf3b168e8048cd7b03c9b4d56d072214ef85

components/libwally-core/upstream/src/secp256k1/src/modules/ecdsa_adaptor/dleq_impl.h	6275	d5973e606b4225b4cbce6ea02496385f3a8e9
components/libwally-core/upstream/src/secp256k1/src/modules/ecdsa_adaptor/main_impl.h	15004	157bf8e45ad426f56583b7def22d91418c0d4
components/libwally-core/upstream/src/secp256k1/src/modules/ecdsa_s2c/main_impl.h	8784	20bea71274c43c26093e5e00a22a9a0ac97ec
components/libwally-core/upstream/src/secp256k1/src/modules/ellswift/bench_impl.h	4399	8973d64bda27eb72a4807aae665a1a2932d62
components/libwally-core/upstream/src/secp256k1/src/modules/ellswift/main_impl.h	27602	8deffa7b37a0602be2b87d0378419410e6cc9
components/libwally-core/upstream/src/secp256k1/src/modules/extrakeys/main_impl.h	9961	a58762afb6a494aaec984d2e65adacf1125be
components/libwally-core/upstream/src/secp256k1/src/modules/generator/main_impl.h	17287	756c2aea7b7221d6b523ae17d367a6e819b67
components/libwally-core/upstream/src/secp256k1/src/modules/generator/pedersen.h	904	6529271ae3f7e786df540f650164e66cae354
components/libwally-core/upstream/src/secp256k1/src/modules/generator/pedersen_impl.h	1729	1cfde5758fb1d3cbe4a2b8f6e9f6ef00eeb75
components/libwally-core/upstream/src/secp256k1/src/modules/musig/adaptor_impl.h	3359	e22baaaa040a80c84f0cd72a8aef3b6ba4c26
components/libwally-core/upstream/src/secp256k1/src/modules/musig/keyagg.h	1324	5349443b84525a85434ff8f92a1e7542c0982
components/libwally-core/upstream/src/secp256k1/src/modules/musig/keyagg_impl.h	11171	fa67be67a1cdb0ad05fc6fd5fce158731ea49
components/libwally-core/upstream/src/secp256k1/src/modules/musig/main_impl.h	452	ea8d83ca27ceafc2d119c82d6297584030287
components/libwally-core/upstream/src/secp256k1/src/modules/musig/session.h	861	c256c3742a68849520569845640f152a22a4a
components/libwally-core/upstream/src/secp256k1/src/modules/musig/session_impl.h	31530	ddcb0d007ebaa3ccf9611de3671c8551ef93e
components/libwally-core/upstream/src/secp256k1/src/modules/musig/vectors.h	37797	c7cd9dc3804d864d17e43c5b00335a405a3bc
components/libwally-core/upstream/src/secp256k1/src/modules/rangeproof/borromean.h	1167	204588b38d00b1db3765ab4b2c25e01f5afb5
components/libwally-core/upstream/src/secp256k1/src/modules/rangeproof/borromean_impl.h	7977	4e1e6b782851715a87440d62bbfbf7f6db6d2
components/libwally-core/upstream/src/secp256k1/src/modules/rangeproof/main_impl.h	5051	5403c2086d4b81f9e85a91f2290cafd3234e1
components/libwally-core/upstream/src/secp256k1/src/modules/rangeproof/rangeproof.h	956	a8d6285ada07204cc46dab26087b3c61b732c

components/libwally-core/upstream/src/secp256k1/src/modules/rangeproof/rangeproof_impl.h	26166	94685312146bc3ac7ead5f4a29cddf7d9c51b
components/libwally-core/upstream/src/secp256k1/src/modules/recovery/bench_impl.h	2320	20d13192a4aabaa52ac4c2a2da772d8aafef0
components/libwally-core/upstream/src/secp256k1/src/modules/recovery/main_impl.h	6153	93fb7455adaa364a3c1f8acef82ba409b88a4
components/libwally-core/upstream/src/secp256k1/src/modules/schnorrsig/bench_impl.h	3795	9655992c7fd214231623716a738a1928e24a8
components/libwally-core/upstream/src/secp256k1/src/modules/schnorrsig/main_impl.h	10970	138c330969201f4d7ff5eb512715b7e05665c
components/libwally-core/upstream/src/secp256k1/src/modules/schnorrsig_halfagg/main_impl.h	7991	ba7c408bf639159630920933e29fb97be7be5
components/libwally-core/upstream/src/secp256k1/src/modules/surjection/main_impl.h	16513	5a8dc46515e1a885f221fc8c036ef08c219a2
components/libwally-core/upstream/src/secp256k1/src/modules/surjection/surjection.h	1066	9ed963634320ce5779dfadc4543cea1549372
components/libwally-core/upstream/src/secp256k1/src/modules/surjection/surjection_impl.h	3758	96034d371d002e572e6e3e7b79aadf3652f13
components/libwally-core/upstream/src/secp256k1/src/modules/whitelist/main_impl.h	6398	a6cb604aa66933d4c5bc1b8e9942eb2c8ea8f
components/libwally-core/upstream/src/secp256k1/src/modules/whitelist/whitelist_impl.h	4653	6d5bfd9b6dde5d1186c0c886f3a8935b8f30c
components/libwally-core/upstream/src/secp256k1/src/precompute_ecmult.c	4017	8085ea86390eab2afeeec8b4f952036492fb8
components/libwally-core/upstream/src/secp256k1/src/precompute_ecmult_gen.c	3864	23d1dadb40296d2752d024678e405b18d5c90
components/libwally-core/upstream/src/secp256k1/src/precomputed_ecmult.c	2409168	1fba19c482de09f566ed9e7bf6a526e4edb59
components/libwally-core/upstream/src/secp256k1/src/precomputed_ecmult.h	1585	d792b8a49759deff5f1e4092558fa9ae11d24
components/libwally-core/upstream/src/secp256k1/src/precomputed_ecmult_gen.c	259715	c0a7af0884b7b07962be67e9da674e0d75a8c
components/libwally-core/upstream/src/secp256k1/src/precomputed_ecmult_gen.h	973	8f514bed441bccf7e34ff05ff49d3a44e9395
components/libwally-core/upstream/src/secp256k1/src/scalar.h	5729	87fea4d072b4db4a093527e7df75b654e66cc
components/libwally-core/upstream/src/secp256k1/src/scalar_4x64.h	801	42aed583b28bda7982c54a6f63e50c0ae5f9c
components/libwally-core/upstream/src/secp256k1/src/scalar_4x64_impl.h	38294	61d0fb4bf51c36819bb778479e7735605d64f
components/libwally-core/upstream/src/secp256k1/src/scalar_8x32.h	725	0614cea113aea88a47ef3b2ba8167722e850c
components/libwally-core/upstream/src/secp256k1/src/scalar_8x32_impl.h	32526	2fe7d67f3218e9fd60fc959a302571e7fbed0

components/libwally-core/upstream/src/secp256k1/src/scalar_impl.h	12113	6ba654bd4edc0349d1d7f1dfc25ba2d73dd9f
components/libwally-core/upstream/src/secp256k1/src/scalar_low.h	1135	3cab20db10224ff85ee7c6d975d2968c1370e
components/libwally-core/upstream/src/secp256k1/src/scalar_low_impl.h	6258	c7c73a26dae95abd861b7927e5c0a1add5dd7
components/libwally-core/upstream/src/secp256k1/src/scratch.h	2224	4af5e1ff665c0418bcc5b1c2de954e7b377f9
components/libwally-core/upstream/src/secp256k1/src/scratch_impl.h	3677	294a55d429fa4cc9dec46ade345bbb2bfdff4
components/libwally-core/upstream/src/secp256k1/src/secp256k1.c	34307	825b8e74f872f89ebd4d7bd3e404a889cf9e0
components/libwally-core/upstream/src/secp256k1/src/selftest.h	1494	d695cd5a7369b372285ba0371edb6bebdb3dc
components/libwally-core/upstream/src/secp256k1/src/testrand.h	1950	bfe1b9a78209aa90d5652a83efccf633bbd87
components/libwally-core/upstream/src/secp256k1/src/testrand_impl.h	5786	ac12496cdb04f5432b14dbb60ab4643176695
components/libwally-core/upstream/src/secp256k1/src/testutil.h	4945	5a780fc16fcd8549ea4774d15beb2d36c6ad9
components/libwally-core/upstream/src/secp256k1/src/util.h	16482	a203e19a64dc197ac051aadad15163515fb01
components/libwally-core/upstream/src/secp256k1/src/util_local_visibility.h	402	d47d4a58f40157ab2a298151f366c5ebec8f7
components/libwally-core/upstream/src/secp256k1/src/wycheproof/ecdsa_secp256k1_sha256_bitcoin_test.json	299012	1be8742064fec73d670339f0036dec56b21ba
components/libwally-core/upstream/src/secp256k1/tools/symbol-check.py	2416	1706069df728c1a2ac1d2578897076713c16e
components/libwally-core/upstream/src/secp256k1/tools/wycheproof_utils.py	427	9f2c2368953eeb3c9b7c3b31a807cde76b310
components/libwally-core/upstream/src/sign.c	28672	9e52718e9015f768784b457a4401d17c9d520
components/libwally-core/upstream/src/swig_python/contrib/aes.py	2010	a908a69ec88322237a640666b30549f7ffbcf
components/libwally-core/upstream/src/swig_python/contrib/bip32.py	2642	3f656b4c9cf631614cc98b3bd4fb86e83d25e
components/libwally-core/upstream/src/swig_python/contrib/coinselection.py	865	e984a050ea05ed45367d0b9beeb21b25a0395
components/libwally-core/upstream/src/swig_python/contrib/descriptor.py	1238	293f13173b5700a5670f3fa59c09320769473
components/libwally-core/upstream/src/swig_python/contrib/elements_tx.py	8121	f7ed910ed53a4ba8f0cbc4cf65f2f998b8753
components/libwally-core/upstream/src/swig_python/contrib/mnemonic.py	1913	0affbe17869a1d77bc388af7b61d71b7eadde
components/libwally-core/upstream/src/swig_python/contrib/psbt.py	42984	de4eede31606e8bf3e7bf885648765a1214fc
components/libwally-core/upstream/src/swig_python/contrib/reconcile_sigs.py	2854	0160b8f879594d0c33c9e08e0e0699e3d65a7
components/libwally-core/upstream/src/swig_python/contrib/sha.py	2019	3a36b14987e4051ba544d89424b76f37bede9
components/libwally-core/upstream/src/swig_python/contrib/signmessage.py	1589	62c57fe76135de42dcd48c08bd33a0165ec56
components/libwally-core/upstream/src/swig_python/contrib/tx.py	4916	23ea1bea593ab10be4a4ade0c979ddf184b57

components/libwally-core/upstream/src/symmetric.c	1767	746f5c8d602544e0db731e54a287703536d6b
components/libwally-core/upstream/src/transaction.c	124444	c9e1504610735267d7048c56d49da1935a29c
components/libwally-core/upstream/src/tx_io.c	38747	a3d59555fa1f81286d92027853842c8e1d094
components/libwally-core/upstream/src/tx_io.h	719	9d25152580f4b1d41d7e6f6b372a72b2e1df8
components/libwally-core/upstream/src/wasm_package/package-lock.json	126902	7f38f77a6cff50d36c4be0c6c88df743fc3ef
components/libwally-core/upstream/src/wasm_package/package.json	608	f1c0400fbd6b50d7866ede0d6c960bacfaefc
components/libwally-core/upstream/src/wasm_package/src/browser/assert.js	321	67e799d7b585943af24ff3d31d5d25b556d3f
components/libwally-core/upstream/src/wasm_package/src/browser/crypto.js	114	1a0550b75fc50491d16e6564434529adaa5e7
components/libwally-core/upstream/src/wasm_package/src/const.js	18459	10b0567bdac844dec0eb80607e7343982967c
components/libwally-core/upstream/src/wasm_package/src/core.js	11103	cdd31896d2aff07dfa172e62b63c0d6c7d68f
components/libwally-core/upstream/src/wasm_package/src/exports.js	110	95b7045e0888c44564dbdad98a4a97d101627
components/libwally-core/upstream/src/wasm_package/src/functions.js	117722	bc0c2b8dad4780ed4a3232db0d0b00976071e
components/libwally-core/upstream/src/wasm_package/src/index.d.ts	94390	e6e48634893b92ee0f07069971ab4d859e019
components/libwally-core/upstream/src/wasm_package/src/index.js	644	e604bf7297be553a02b5a716288bb479d6b39
components/libwally-core/upstream/src/wasm_package/src/util.js	1217	34e03d693697456630eb3a9db4d8f253271b6
components/libwally-core/upstream/src/wasm_package/webpack.config.js	458	9a861d818ed3d7e3b9cae555eb82409c4acc7
components/libwally-core/upstream/src/wif.c	5335	2e01925997bcd34600c7e6e7e8831120b157f
components/libwally-core/upstream/src/wordlist.c	2726	d3f630d36f8a7763a67daa88b7652ed40f775
components/libwally-core/upstream/src/wordlist.h	1678	7e8993598f450d95843cd4217f2d4fcfccf34
components/libwally-core/upstream/tools/build_psbt_ctests.py	854	ecc8332fc3ea0a904897ba05598608e174a64
components/libwally-core/upstream/tools/build_wrappers.py	26927	b59803b7ca9b045961bf6f446dc6c1d24b8c7
components/libwally-core/upstream/tools/wordlist_cc.py	1736	e389277f851b59a3d4167ffec131903087ffe
libjade/daemon.c	13786	f7a879446d031324ab8a7c6f3852fe798423a
libjade/esp_camera.c	4097	2dd75df9a8ea04009e2292489933d9fcf3d34
libjade/esp_event.c	7180	232359c418327dc712fe37fec5925be3f4376
libjade/gui.py	33303	83bac459603739e79701326b723f64519eedf
libjade/include/arch/sys_arch.h	0	e3b0c44298fc1c149afbf4c8996fb92427ae4
libjade/include/driver/gpio.h	0	e3b0c44298fc1c149afbf4c8996fb92427ae4
libjade/include/driver/spi_master.h	0	e3b0c44298fc1c149afbf4c8996fb92427ae4
libjade/include/esp_app_desc.h	199	91d86dc8cff605493b74d674ebbc952013616

libjade/include/esp_app_format.h	110	bdbc9e3ed96e60b8ae18596b13293292637dc
libjade/include/esp_camera.h	2691	dec13aed2560745cced0c231bb50400c53cbf
libjade/include/esp_chip_info.h	210	6a640247301af5c1544d439e9b4fcf26e8cc7
libjade/include/esp_crc.h	223	bfd084c9d6b84484f924da49dd4554838791
libjade/include/esp_efuse.h	0	e3b0c44298fc1c149afbf4c8996fb92427ae4
libjade/include/esp_err.h	345	42989c30b40bc2ce4dc91c1fb3a22d7ab528c
libjade/include/esp_event.h	1178	025b67230b51a1a2a82ca7203f0644fec01a3
libjade/include/esp_heap_caps.h	1073	7b0a2298d9aac0b4f4aab1792c6c2f189e361
libjade/include/esp_idf_version.h	153	6853a6164c0590cfea0ccba19e580a18ca0c4
libjade/include/esp_log.h	2829	536f591ec2787696398452c2ede62f0a325ec
libjade/include/esp_mac.h	163	8b3ef41835b640d41502fcaf251b127737cdc
libjade/include/esp_ota_ops.h	1359	ac9147f3061e1552a023fb091f98a1fea94ef
libjade/include/esp_partition.h	603	ba52b5f0a106e4fe45c6a6d40b0a432e9bce9
libjade/include/esp_sleep.h	129	714e5d29e39de498e4e5fea7dfd7d48971c96
libjade/include/esp_system.h	119	7433510b853e2aee76e094c0777431eefa692
libjade/include/freertos/FreeRTOS.h	173	d396cdcc4f08e165f974bf7a7531e645e4099
libjade/include/freertos/idf_additions.h	399	1421e9defdcbdeab90ae75a5930d5b8dc3ba0
libjade/include/freertos/projdefs.h	260	39375a0187d7e96877292534c011376a07c93
libjade/include/freertos/ringbuf.h	3484	5372dd67671bab23bede2abb75bd29ea496d0
libjade/include/freertos/semphr.h	3637	e4e90504e643d2a7ab1781a0a9968e02320ac
libjade/include/freertos/semphr_darwin.h	4571	63e6beea3c49daa4ad8d3ea36f710a4ae5e14
libjade/include/freertos/task.h	1443	cfefd801a2183bf91518cb31c05c14a27f674
libjade/include/freertos/timecvt.h	1278	3ce86cfd1cf226da9a45b94a52eb287773ac2
libjade/include/jlocale.h	133	d082d6fc1f54bb2e4293df4c07bf2b913266c
libjade/include/libjade_port.h	1472	ec5e392b780f7842c48506bb308fc9e47acb3
libjade/include/nvs.h	92	fea28e5ca7d2fe5973f6e8cefdc982958f40f
libjade/include/nvs_flash.h	1999	a155f6dfec622f88f08939da808fd662cf8b0
libjade/include/sdkconfig.h	1133	44acd257bf5c3bd6c5237a85ab0c73b4b789c
libjade/include/sodium/crypto_verify_32.h	250	044b7a02852f333fdc1db96edb29747cf99f8
libjade/include/sodium/crypto_verify_64.h	250	a8d8acf4c3fe4df0d2bb8660fb8fa20be40ce
libjade/include/sodium/utils.h	210	7ef7d329ae47eb7fab7e6bc8d76c533c2e806
libjade/libjade.c	15927	804d9d5bb845f69067b2a3e393cc605031f40
libjade/libjade.h	1401	cebd1bc02d72cc810cb7e0f372006b199812c

libjade/make_icons.py	440	3cfbc4d0df1e563be89fc43e1781e1340f6bb
libjade/miniz.c	94568	9b6ea606485c1ce9e1155b11e3d86710a4fde
libjade/miniz.h	33110	2a2c531f30ae3d6611c8956135a6d6cb8150a
libjade/nvs_flash.c	8819	2dab1ea45e332ceeebeb1fead3d222667a9ce
libjade/selfcheck/assets.c	3680	b6d0fcf0640be91826ead9d811ab732f680c4
libjade/selfcheck/bcur.c	18883	f240136dad29434b390d8e214d682ac5ed73c
libjade/selfcheck/cbor.c	2740	f9e6cc1b46f0669504c55bb2b84d0fda74804
libjade/selfcheck/descriptor.c	23625	8fb673c7ba9825eb07d5a3c804c0320111726
libjade/selfcheck/task.c	3777	2cd922f9d11b3d1e35e4b1cd7086179fa3062
libjade/selfcheck/urldecode.c	9085	952e3646b3aa5cfb9876921eaf6746839597f
libjade/task.c	14246	cffbe18f86ce557fe1fd87ad5c709ccbcbf98
libjade/tcp_bridge.py	3297	f53615430e1d62c23859662b383eaf4426dd1
main/aes.c	2577	cc9a126c7be550237fbd6b27f715a486e96a3
main/aes.h	802	49bec702f79dbd7d5d4a0095b126c26314441
main/amalgamated.c	5318	5e3b8981f49e16b1b72264eb4ca9a2c1f3d24
main/assets.c	10345	884bcab009426af20ebe5cdbcc363a89e7d30
main/assets.h	1071	a80600f9c7f8fad97e9278a76eb507d70095e
main/attestation/attestation.c	29559	109fb6032a9e94ab16af11fbdcaab65966f8b
main/attestation/attestation.h	1262	62b25a73c9ccfe3671d11c836620bbc9c1098
main/bcur.c	33709	71c649ab160724204b8942429bbebd8f596db
main/bcur.h	3404	7365bb4b7cd9a9dded0662f1d560568c6189f
main/ble/ble.c	33067	e1f41bac969eb61fdd048f1c67a5e8d0a8973
main/ble/ble.h	846	57357773fe66db5f2e5cbc19a7f7402386418
main/button_events.h	7009	72f3d0a7a66c62d5b6bb47a9fd8262302cb1e
main/camera.c	26492	5980e68ee7592f58aaf481ade8a80fab7ba89
main/camera.h	2229	5bd4c26b6cabf613409bcbf5928a1c60acf91
main/descriptor.c	30078	81985445c447d2314ad3fd3d3bb35ba19a049
main/descriptor.h	4566	639e67c90b499e92f7f1f88f941953419f03c
main/display.c	31542	1d537d51c1c3f7a7056c86856092f5e84847e
main/display.h	3421	68f42884be36496804252efb8456c7f8f5403
main/display_hw.c	14900	208256db8356a502065fd95d846f3fd59b4ea
main/display_hw.h	508	777d9b714ecc0057fa9e03a17d978dfc4a51f
main/fonts/BigFont.c	19732	663d44895509eb139ffee9384db9e1a3fc4ee

main/fonts/DefaultFont.c	10068	4786af48b658f7e44359236665830c5e4884
main/fonts/DejaVuSans18.c	13737	9e1316d88900d168bef760941843e5e811b5e
main/fonts/DejaVuSans24.c	20500	4151ea519d38c7f95214ec30558f579435d60
main/fonts/Retro8x16.c	10239	f2e52ea1783a24d45af641fc7fe44a0e2d336
main/fonts/Sinclair_M.c	19738	9b9720a19ff7b1dbde6093ce0cfdabb2c017c
main/fonts/Sinclair_S.c	5674	4f0fe35115a3d379c910fef96b5205f1e6721
main/fonts/SmallFont.c	8209	51d60606ec2e687d505314fa6eedc72940e24
main/fonts/Ubuntu16.c	11651	c3c604f1beeb95251063b8c6c16233ae98e4b
main/fonts/Various_Symbols_32x32.c	76386	728647e3f9be98442e85df84c3765cc4368c3
main/fonts/battery_24x48.c	9261	258c579414b1001f2e6f1a9a4783ee8b20107
main/fonts/comic24.c	20269	06eb7dd9c117b59154c500e9d59f33ee7fdd8
main/fonts/def_small.c	8577	d4202e5dfe6c3e69249f7875c03ca85e6a3e3
main/fonts/jade_symbols_16x16.c	7691	c90585ea9df24bc4d562a85cf2ab6ec07a693
main/fonts/jade_symbols_16x32.c	8452	44d914a4371de44e0e50e9435ac01093f86d7
main/fonts/jade_symbols_24x24.c	3381	803ae20ecbdeb75d86b4976abcf17a5412435
main/fonts/minya24.c	20253	ab78406f29ede3ea4b2904d32e1b01be5e970
main/fonts/tooney32.c	36755	bbaed9670759c8df1d98ca5f4097773169582
main/fonts/various_symbols.c	19771	e61f4091679fe50badcfd5f6e99652acd4004
main/gui.c	106393	9c7b0771bc38bf80564d562273bb6c1549f30
main/gui.h	16914	31c0755842f1d159f3c08e65ad1d54c1a7a66
main/identity.c	19077	61059fdd5caf7e8236b18424591656cb56821
main/identity.h	2296	cc5c1d0a6f9809e090fda870b683cc6ea0d55
main/idletimer.c	9634	fd3090c7a71f9d4c396baf8b5bac86931969e
main/idletimer.h	249	dc55b5f0d24341f66247ad56f88a45e02eb91
main/input.c	595	abde01fe56c425c7f95cf6017eb870eafbf14
main/input.h	81	25edf47c16e5569a9b7bcea838d2e4371d2d7
main/jade_abort.c	1545	e500f0952405f3443b8ca8d4ebae36137c99e
main/jade_assert.h	7244	8e2f6ec6b883ff3357d04381d912dc06ff4f0
main/jade_log.h	430	031cfd4fe846aa37be2cfeb37ea9584fd93f3
main/jade_tasks.h	1193	2c2b6cbc8004fb5db54eca9a5f82265c228b4
main/jade_wally_verify.h	693	aa4dc137a8a11fac416641f3d5525cba6c5ff
main/keychain.c	26182	73e309e0c6c62dbcd90021db04bff0f562910
main/keychain.h	3092	5283bcf2460992938c5e0bd25f88c0cd61b6c

main/logging.c	2831	7681052739d1b4ab49a26eddfba5daaec2e76
main/main.c	10371	ffbeb1491192c7be16cf1ca3f12a1896ac01a
main/multisig.c	27403	ad3b2fe43a0a531aa916a69ae77b378256c2e
main/multisig.h	4197	f1d1cbe6dce6f1dac7c790e02fd272811ca35
main/otpauth.c	30114	da1f2217ba05ef1b5c5b80f155eaf2575e9f8
main/otpauth.h	2538	b28590eb82b20f88f9230141b8e4ac5711c84
main/power.c	1744	c1153c8adaa9dde69ca6206ab6fdb8aa55e3c
main/power.h	907	337a0674fa7a49698cdc64e2e6a035507fcac
main/process/auth_user.c	19586	0379d767bf64c8b99dd28b8f629d515e06331
main/process/dashboard.c	112704	f07a5a347910bbcdd4ec852327906e1c3abce
main/process/debug_clean.c	2765	fbfc93f503c64502461f60e619f233f72422e
main/process/debug_handshake.c	5924	118f6d105d4b762a674d90b8cb31b09828dc1
main/process/debug_scan_qr.c	6890	3a49e44f207a6462197cefc167789de00580e
main/process/debug_set_mnemonic.c	5198	617984672ee4e7323d596062da494fd90c20a
main/process/get_bip85_entropy.c	13579	a6fda19e47b899e0206e5262ea531970ed33c
main/process/get_bip85_pubkey.c	1316	9b919409523ec203f5b65df60a6bed904f4eb
main/process/get_blinding_factor.c	2825	67b1e5ce79c11dcbaf7a2d4965aaa026e3f46
main/process/get_blinding_key.c	1831	3f1109641a7439c8d23fe6571e30269324364
main/process/get_commitments.c	5486	220d67c6e562634b7264d5dd4f1835d70ad3e
main/process/get_identity_pubkey.c	2238	d687e7066ff605b7c37387632c747fd4a968c
main/process/get_identity_shared_key.c	3180	243cd8e3b2c5773f83ae15ffd5e477cc35a8b
main/process/get_master_blinding_key.c	2254	59a95842547b73bb6a559367f7c72b9f7f4f1
main/process/get_otp_code.c	3725	27526eb95637148583a44992dbbff520fb1d0
main/process/get_receive_address.c	12607	a10e79d45374e7c4564e0fa6398850b7229e6
main/process/get_registered_descriptor.c	4059	4afc6b5ed7a4c9efd4f9e6e7da8aca67ad916
main/process/get_registered_descriptors.c	3917	f533398692a49681212456ecc360b6ae1ae61
main/process/get_registered_multisig.c	7658	dd8d7a58913988701271dbe920106b7685c23
main/process/get_registered_multisigs.c	5150	001f918539dbda9591cf710ebf25b5552ff38
main/process/get_shared_nonce.c	4626	d4280140163dc732d3a51566944e9e65d6d34
main/process/get_xpubs.c	1550	da4030c24a625538b79fbe02cf02f584bdecd
main/process/mnemonic.c	75565	d36ee56949ee52422b148685d056be9f673af
main/process/ota.c	3410	de6f4e915d4316a3569fe3288dae1d5e97a8a
main/process/ota_defines.h	2417	8e6f1399e10a7626e527b09dbb1f2cfb81de8

main/process/ota_delta.c	7711	193538cdbdb3a1d85577f534a8a59ba7e343a
main/process/ota_util.c	18857	c45d9bb4b24c5c35c22918e763bbd41a83a34
main/process/ota_util.h	2644	098bd34e48b2e6bc75fd90a5df8a26d7d6572
main/process/pinclient.c	24785	541c57ef78078aeb565636c58b6e00abf8f51
main/process/process_utils.c	22504	71028ab9c16eca06604bf53d2fba1489f6b3c
main/process/process_utils.h	7881	1d220d075567ef7dd43d1fbb3d8a457e2efef
main/process/register_attestation.c	3488	6e292d309d56a078aa7eff0a7ebfa013c554b
main/process/register_descriptor.c	9695	9d83a4f3d94cc7b8c6230c5544f77b25ccc78
main/process/register_multisig.c	28846	e0048239f36c3e503bcbc622c5f3cecd357f2
main/process/register_otp.c	16858	fea7399f1d0c4714cfdcfd57ecf6f91c947d6
main/process/sign_attestation.c	3898	b7e02234b2176ecd30479ea85d1961aa500fc
main/process/sign_bip85_digest.c	5513	40cf03361d9c783feefd3ce1293e99b27363c
main/process/sign_identity.c	4081	3beb49072c68ae6bde613d5132e07ae256118
main/process/sign_message.c	12825	7d19e713fe281f9c72d17648cc6ab532c4202
main/process/sign_psbt.c	58170	9c1b424fa472cec18cbd1293932e759a7eb5f
main/process/sign_tx.c	45775	7ce145ef4faa8a1ba4eee2fb3027a280dfc4e
main/process/sign_utils.c	33877	179c9d7594218ecc0ac5676c453ac903254c5
main/process/sign_utils.h	3431	fac558fd131f0dd75394e7afa80c42c9c8db3
main/process/update_pinserver.c	11269	207629fe55e21ee8008689f6dff4b19e854f2
main/process.c	27817	5b14f4bff63c5fe0bbd8748d3dee896f1bb60
main/process.h	5262	fef0b8a37b02364b6c4998ad4c38bd31cc8a2
main/qemu/qemu_display.c	9905	41be8d08db80de59b284c69a5f37f8638297a
main/qemu/qemu_display.h	279	2d18a4a60630d521c8f1f39084c9fa2b59ec8
main/qemu/qemu_tcp.c	8461	467b0d175b584da39c4c53e862f0e4ee3601a
main/qemu/qemu_tcp.h	190	fb748cc0decd96ffdbe5306ebbe6fd96f1fec
main/qrcode.c	37506	ad8b8e9ca269cbb0af15a671b42a2ed355c09
main/qrcode.h	3061	8d050b24958572ee090ae3b5bc296d676f500
main/qrmode.c	74171	ae9e0a93c4850c1252e5bdc7908f90d7cd3a4
main/qrmode.h	1820	3400a52b8c0c3efc945340f2321be7885fcbe
main/qrscan.c	7940	91e0aaed8a64ecb703ce9ed9e188cb5397f4f
main/qrscan.h	1702	39ded29b251021ccdee9ac80d99e830dd9f1c
main/random.c	11518	f6ba8919a0bd39348a8964c156aeb20fd9671
main/random.h	606	da655b362510104a23d8406d7087c0cf4fb10

main/rsa.c	5370	44f71d31a8560a1d044784e8b04575fc406e8
main/rsa.h	962	f395fe5d556a96cea8d327ed6e638a6aca38a
main/selfcheck.c	17487	92a640e51db692fe50aaa3b8f0325ffd4004c
main/selfcheck.h	1919	3c7dd66cc36179a72dcff7b702c04660cf8c6
main/sensitive.c	5212	3a765b45b6790fe7860ddaea4c88c02ee9fe6
main/sensitive.h	2101	3a63596396c06650a3a12505730e648062457
main/serial.c	10406	7accefddd676dcb18e28812402429a8f51b32
main/serial.h	309	73bc6fa520e77691571433554ed7fec5f3215
main/signer.c	3883	4172e6d05758443378845b24416de4a537d13
main/signer.h	1452	75068372b5e1f0ec4e0f28591da5f056f11fa
main/smoketest.c	18884	bec571bafd568d90dbf1e75acfe38370774ef
main/smoketest.h	120	ecd8e7374234f0dea70274d6353cf3a9b5dbc
main/storage.c	32538	be9475e8734a339a1a7fccbaa976ce637a6ee
main/storage.h	4897	c768e8fd83e0e62186b4f22d77ad6847eed36
main/ui/ble_confirm.c	766	6c7f6038060d27cb9198593d597484620fe65
main/ui/camera.c	2741	da019d3f1f2f451de4d1eadeaa14222bb3358
main/ui/confirm_address.c	6257	4cd1c91d15e27dce32aeb4b3d4142be4b53dc
main/ui/dashboard.c	38377	467214e13c102a6442edb86909c1aae99c527
main/ui/descriptor.c	15828	eb1b0346d3c8c542344e37514e12a9e6b3330
main/ui/dialogs.c	30011	50251b2002baa585763fc0374e3f048093917
main/ui/digit_entry.c	14903	75d3bd468463dd1771b8113f6f7cd801e28f3
main/ui/keyboard.c	10747	ba415a1f5741ba75a36f97d77d110ea8bb671
main/ui/mnemonic.c	22270	1b4b58645cf10ac3cdaa20778c89f9b32114f
main/ui/multisig.c	15919	478de0b242ddb53846da9309a4541dec4c1d7
main/ui/ota.c	5153	af44d015d6f08b73037bf8e8b2dd92c1df97a
main/ui/otpauth.c	13377	0c6fe581862a47c316b2ad8d6d1b64ca3011b
main/ui/qrmode.c	17928	dbd0b242f964aa6009b34b5191817ecf15f6b
main/ui/select_registered_wallet.c	3457	5d083eac87b93b8b92f893b573e09b4b20bd2
main/ui/sign_identity.c	2408	83319e8311926c95451e83ab6887d65d9d8ac
main/ui/sign_message.c	7103	1c44ed34d71f18a8ec8b555d15455cc0ec236
main/ui/sign_tx.c	31154	c68ac7f25f690a17ab4b665975c2232aa4b6e
main/ui/sign_tx.h	1074	ecb8239a11d8fab093f7397fff4a41592bdd3
main/ui/signer.c	8232	2e59e02e306f4a61d16ef00ff97f3f86ae8a9

main/ui/update_pinserver.c	8359	04de1af452d4a8b6f038f3760887165ef4732
main/ui.h	7837	f2a5526436add3772cd3afc50e062d95641fb
main/usbhmsc/usbhmsc.c	11342	f751c399bd28431b90bcf28cd9f8faab6a6ba
main/usbhmsc/usbhmsc.h	643	64e8a037749b423bf7a8c1cfff1a7d2cbfefe
main/usbhmsc/usbmode.c	36491	cf8387f59fced846d769e6a5cd7dd1e21fe92
main/usbhmsc/usbmode.h	377	47bf18c0bacf8d807a0efccbadabea74d63b4
main/utils/address.c	10283	678a7dd6f33487fd2464125cabe177ca3f7d6
main/utils/address.h	1119	ff85133ff71681e53e70cd986cb5abf570aba
main/utils/cbor_rpc.c	21678	3639ad7f221082f17b991c7ce5df3f15d1290
main/utils/cbor_rpc.h	4584	a8f950b65aa6d1f565eb5df2994bde2ca453c
main/utils/event.c	4978	a486786db4dbf40be24794b2beb072b716ff8
main/utils/event.h	1455	3eb991ef183a40c8ee46564179061b020ecda
main/utils/malloc_ext.h	3630	f276221af422ef5c8e55f92d6f97f4ab5f04c
main/utils/network.c	11202	a645709fe4bc725088719b3d3a3c02f9d4bde
main/utils/network.h	1764	8f099fadd87d511a2075dc1baaf5ef0e63512
main/utils/psbt.c	9261	7e9e009a4aef8329efe88f6f1bc852d869a49
main/utils/psbt.h	2602	f4c081b54cd185597316182accdcf55913c15
main/utils/shake256.c	3236	0f248acd752c2e4ac30b8a7746719b701bfa2
main/utils/shake256.h	406	5d6531b00ecb44ee8d4787238a75d9681c769
main/utils/temporary_stack.c	5015	9225a81af7da8e0fccf84923b54d52a83c36b
main/utils/temporary_stack.h	580	e62674406208b0252fc665066c78f4e05b671
main/utils/urldecode.c	5487	54818da78ccdf14d27f08156aaac6bd7038b6
main/utils/urldecode.h	1694	77a573f188745a53f1ab544ca64adb32c2647
main/utils/util.c	6968	fb9002976cbe5b58c94041f0f9fb97cfc881f
main/utils/util.h	5319	bcaad10af73dc760c8f2d5890ccf824d86a17
main/utils/wally_ext.c	5481	2ed6fa1f3967ba5751a8810051310096851fb
main/utils/wally_ext.h	294	8e4ca78aecbb261fe4401e761aeb3b1365ebb
main/versioninfo.c	6345	8a9fd493300a220db4aca9ea05c39bc71e65f
main/wallet.c	60648	2355de892316eee3d8f799ceec58a988b26ff
main/wallet.h	9463	bcb3ea05f748b4dfebbdea7c59468bb7864e5
main/wifi.c	20043	30820f634e4d57a7e4aa123a8f3e6dd771471
main/wifi.h	460	8273dacaee29f457b4a5453c7ccbbf06b8254
main/wire.c	7543	1a6c534cb518b8698aebbae58c13e7ae41dc7

main/wire.h	746	c7dfa1c20463b7e99997545721fb2aacf08cf